

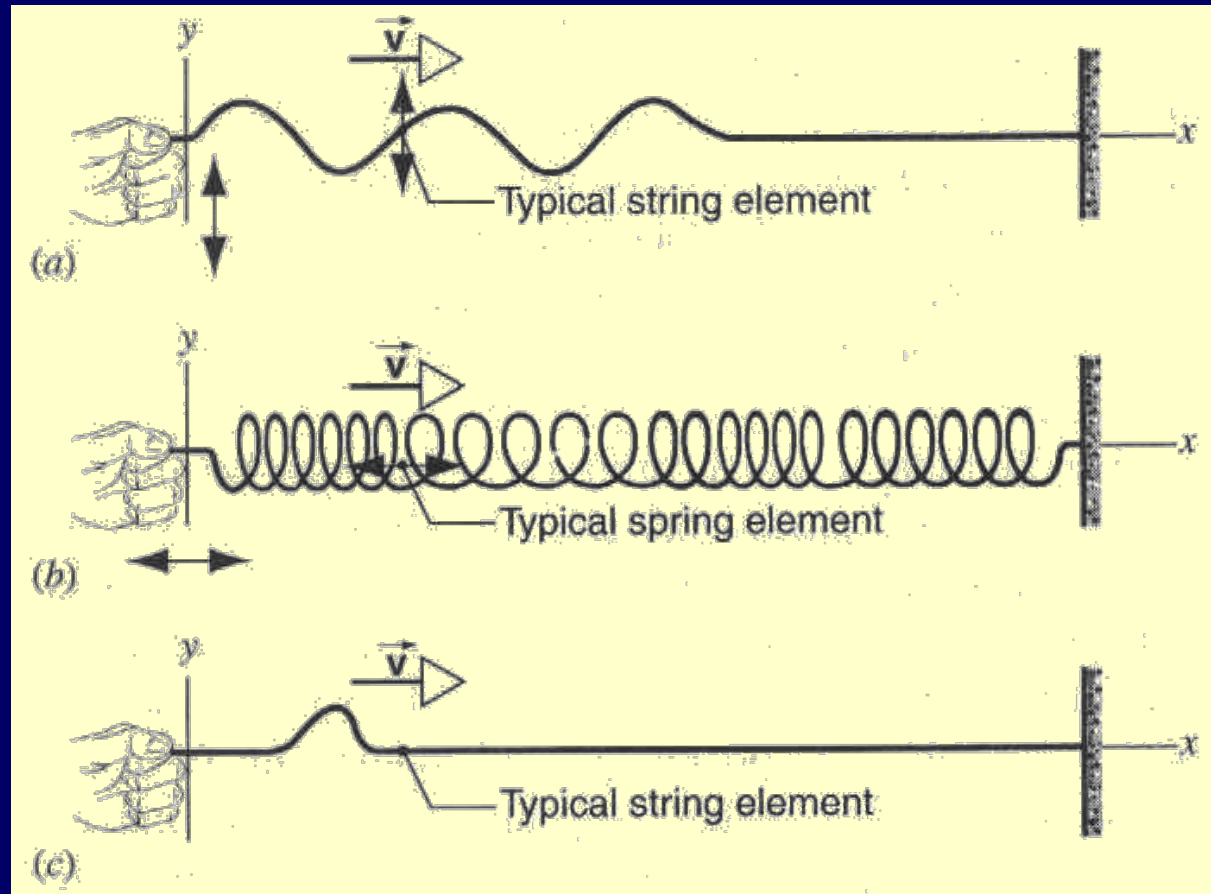
CHAP.18

WAVE MOTION

18-1 Mechanical Waves

Waves are
**common and
essential** part of
our environment

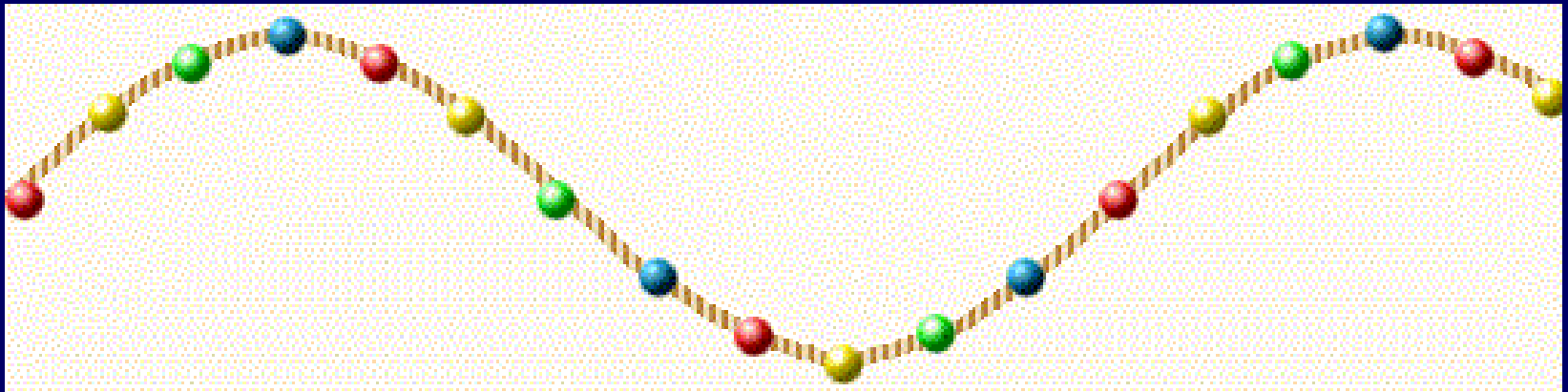
One type of
waves is
mechanical wave



18-2 Type of Waves

1. Direction of particle motion

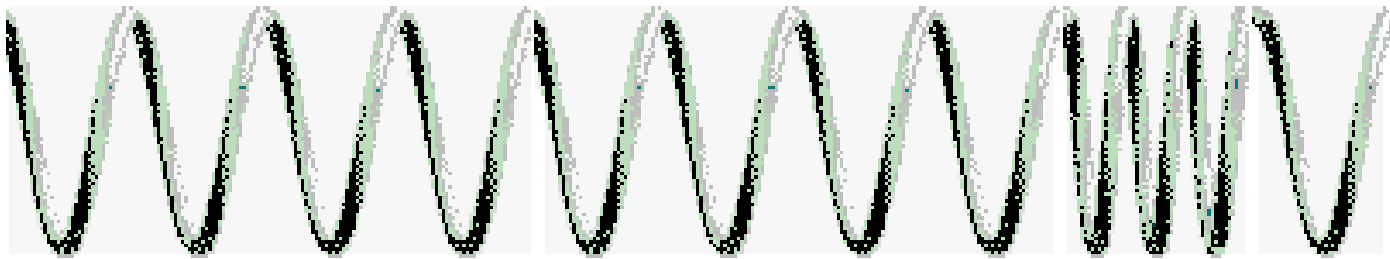
Transverse wave



The direction of motion of the particles of the medium is **perpendicular to** the direction of propagation of the wave itself !

Longitudinal wave

Unregistered HyperCam



The direction of motion of the particles of the medium is **parallel to** the direction of propagation of the wave itself !

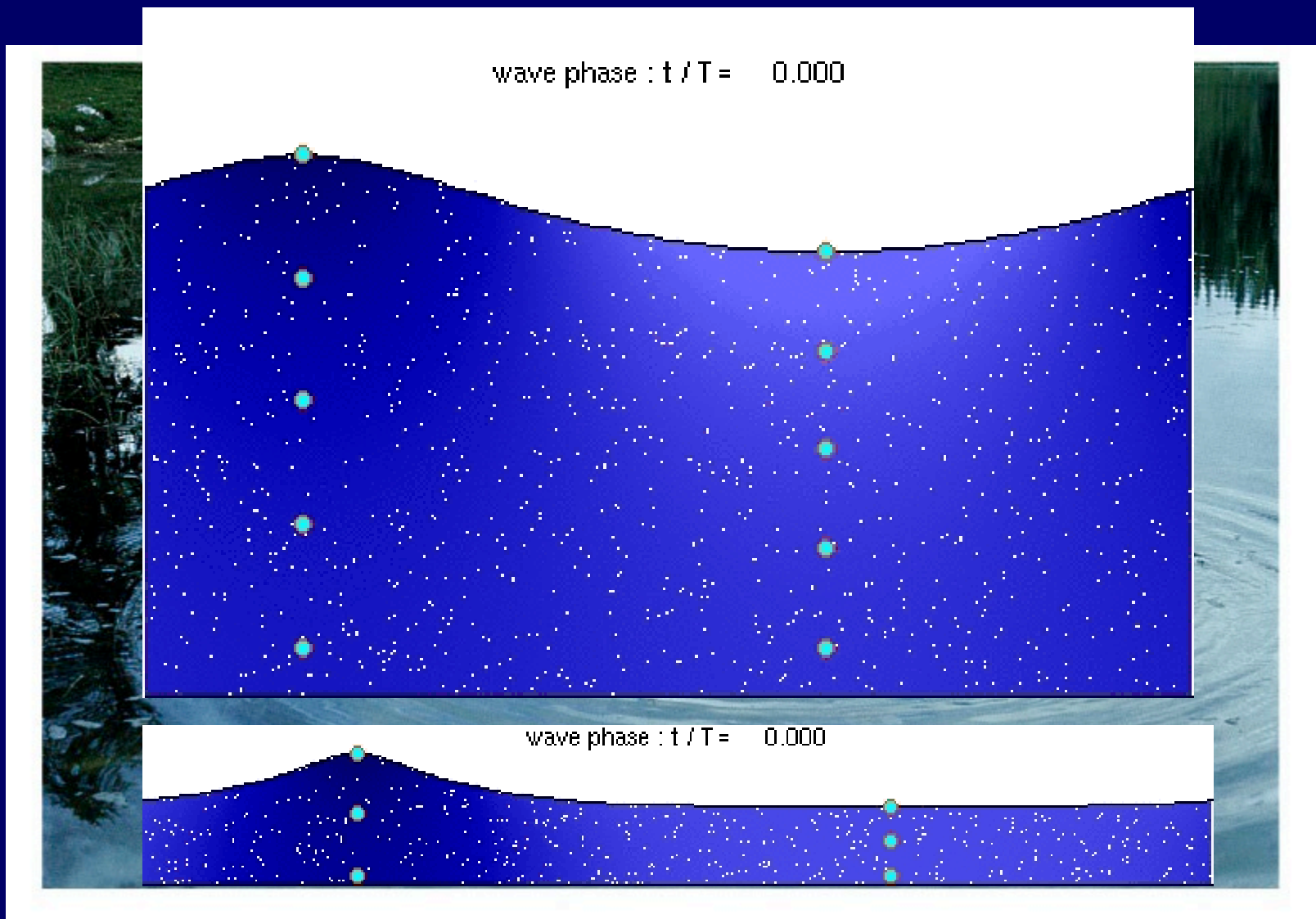


Ocean waves combine properties of both transverse and longitudinal waves. With proper balance and timing, a surfer can capture a wave and take it for a ride.



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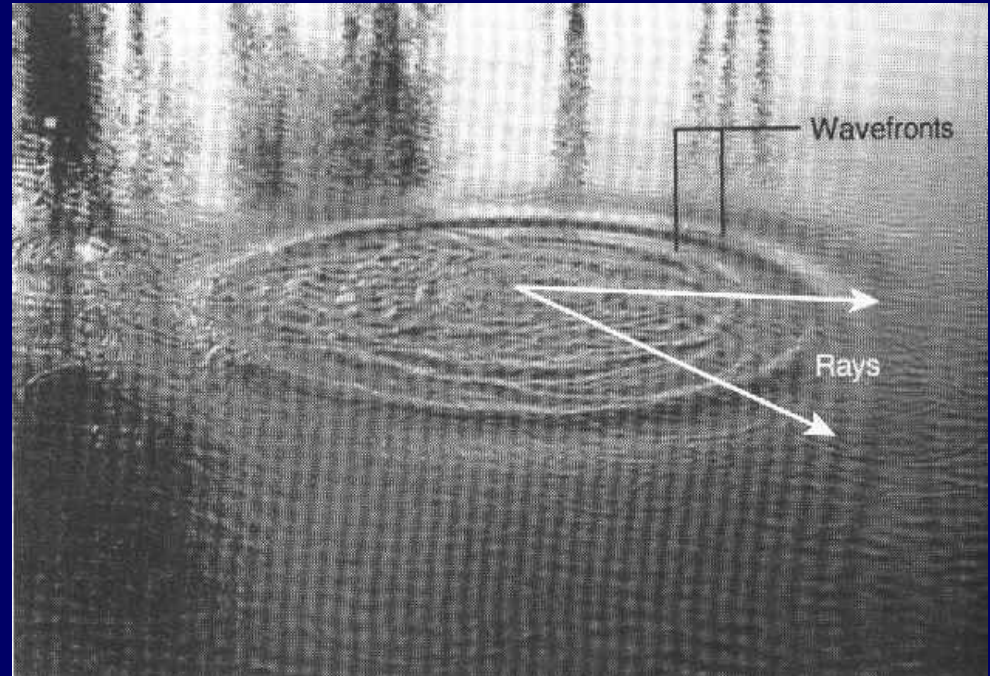
When an earthquake strikes, the perturbations travel through the body of the earth in the form of seismic waves.



Water wave: neither transverse nor longitudinal wave.
There are two types of water wave: deep water wave and
surface water wave

2. Number of dimensions

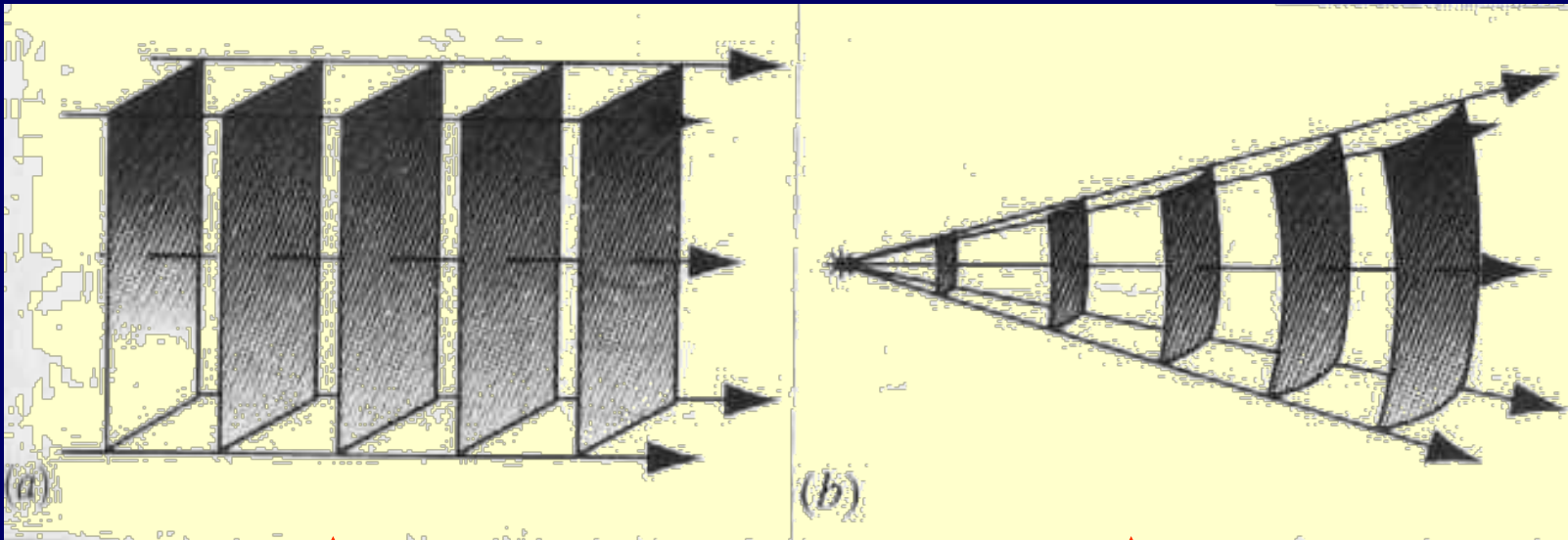
Wave can be classified
as propagating in
one, two, and three
dimensions



3. Periodicity

The simplest periodic wave is a **harmonic wave**, in which
each particle undergoes **simple harmonic motion** !

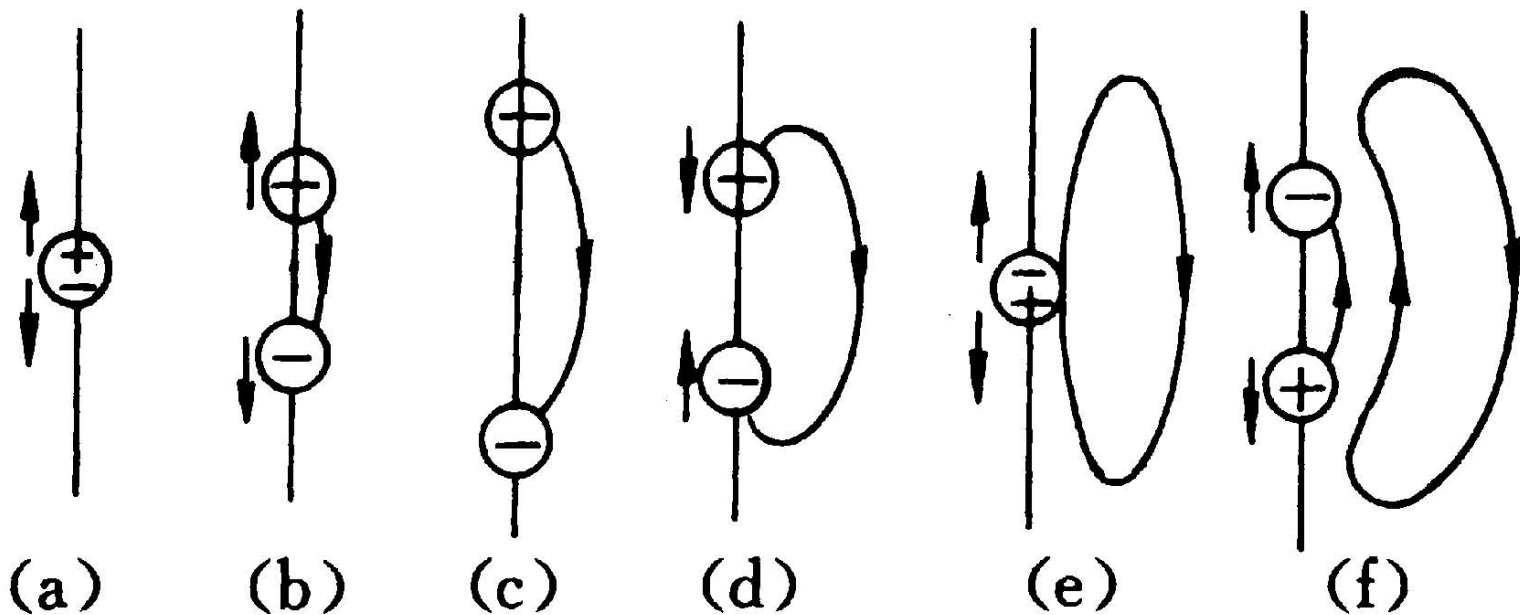
4. Shape of wave-fronts and wave-ray



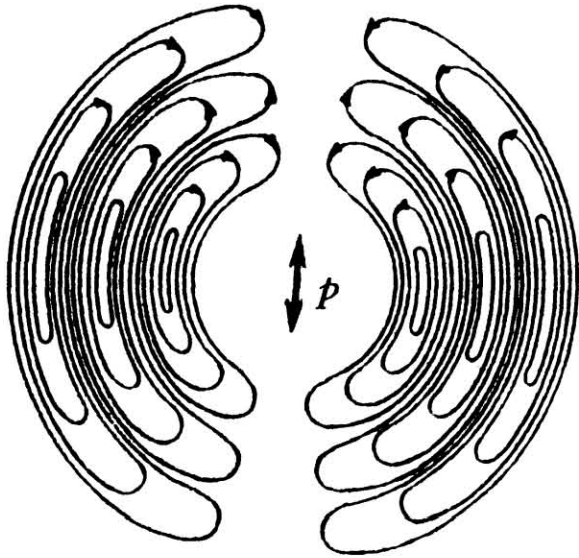
A plane wave

A spherical wave

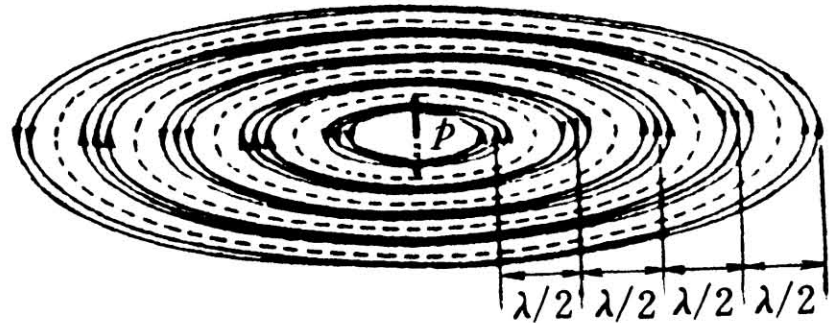
Electromagnetic Wave generated by oscillating dipole



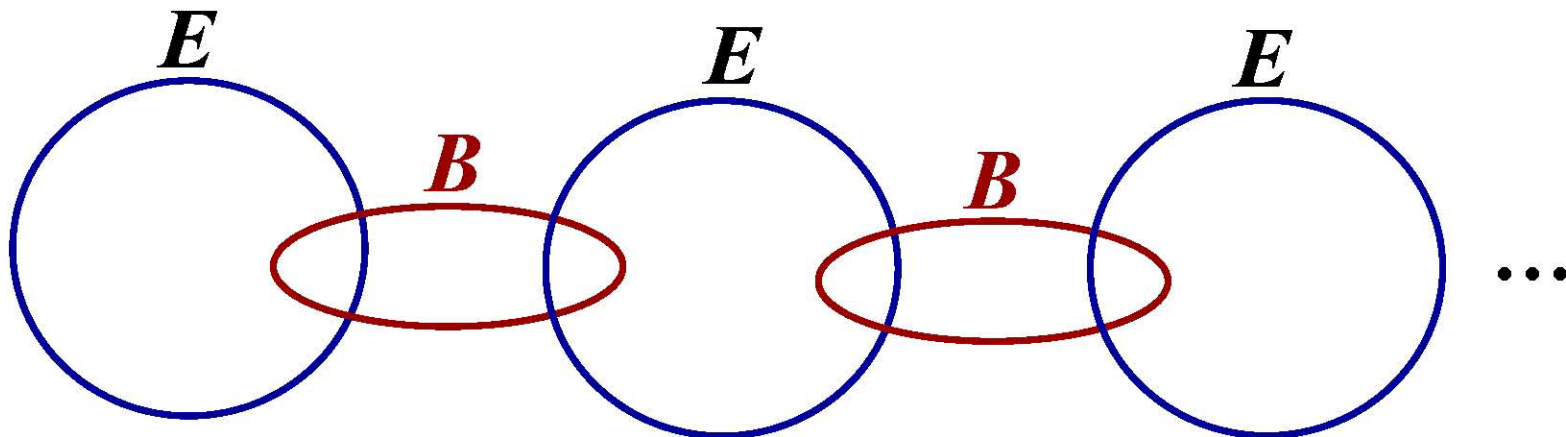
Change of electric field line near the dipole



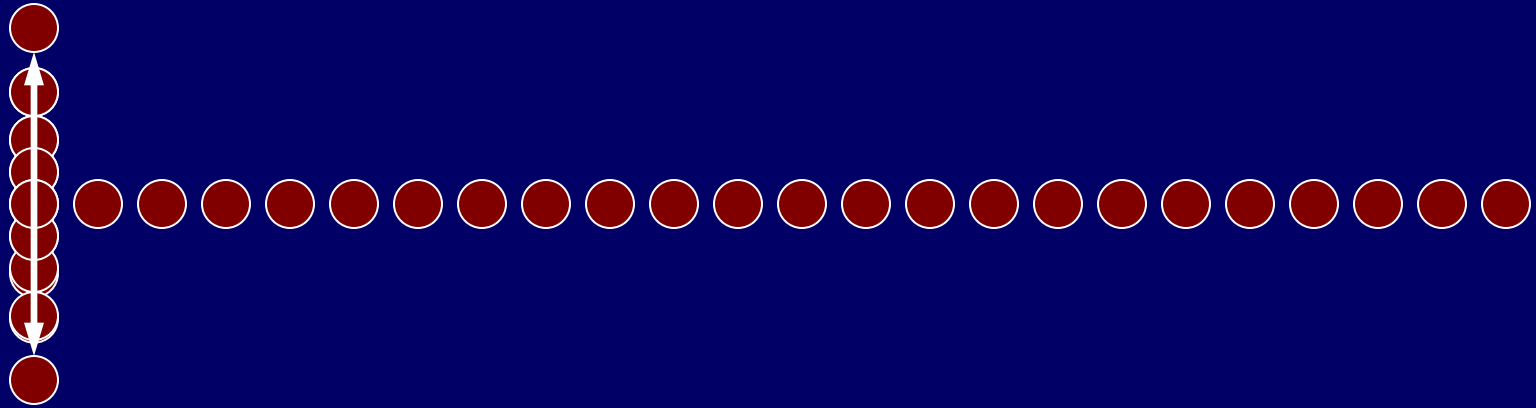
Electric field line



Magnetic field line

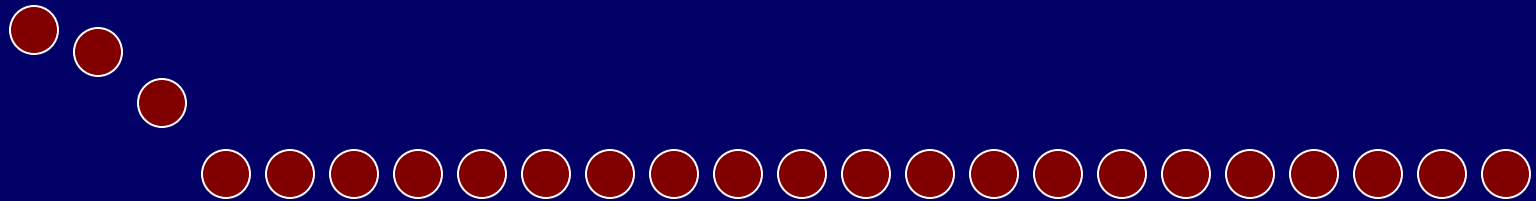


Traveling Wave

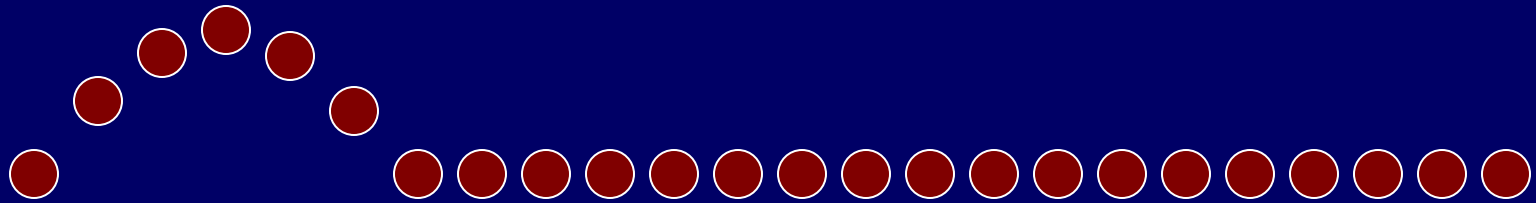


Oscillation

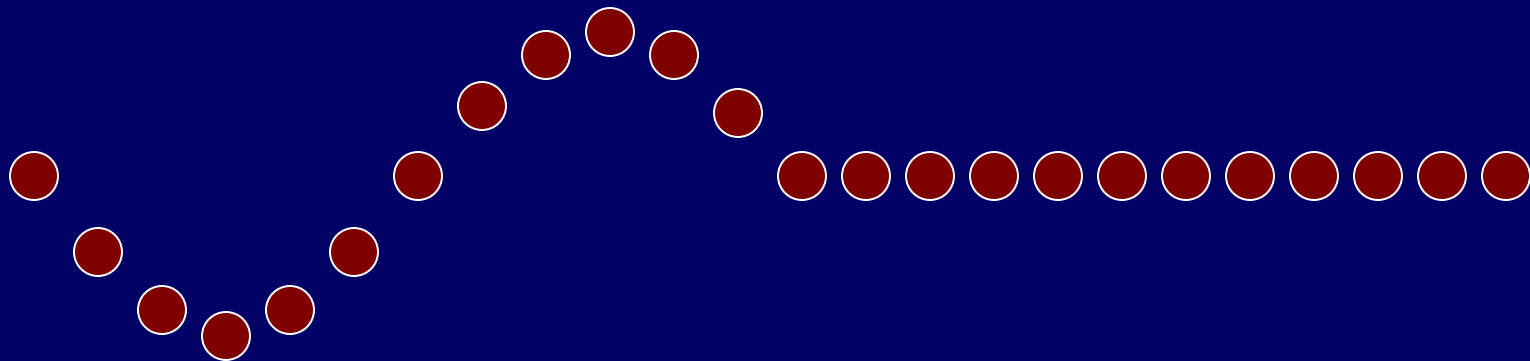
Traveling Wave



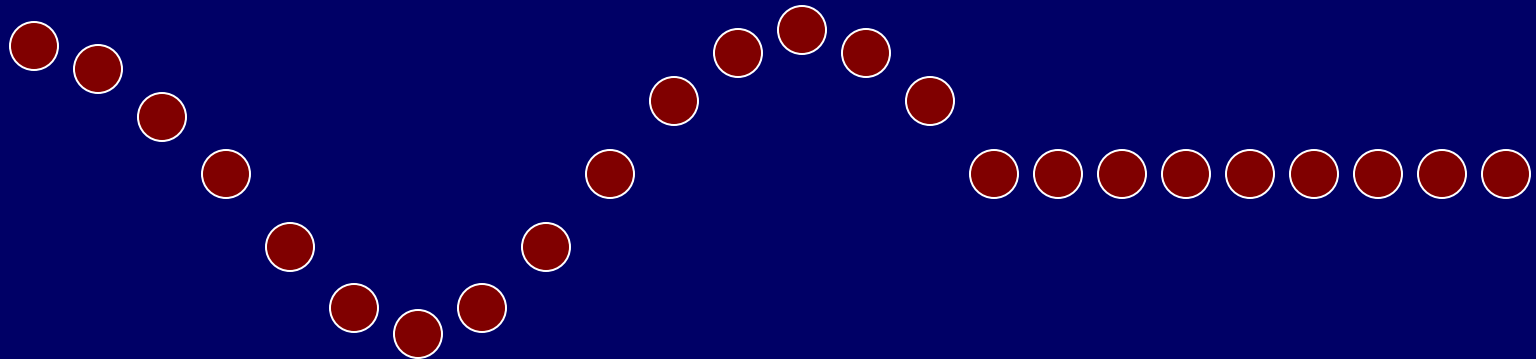
Traveling Wave



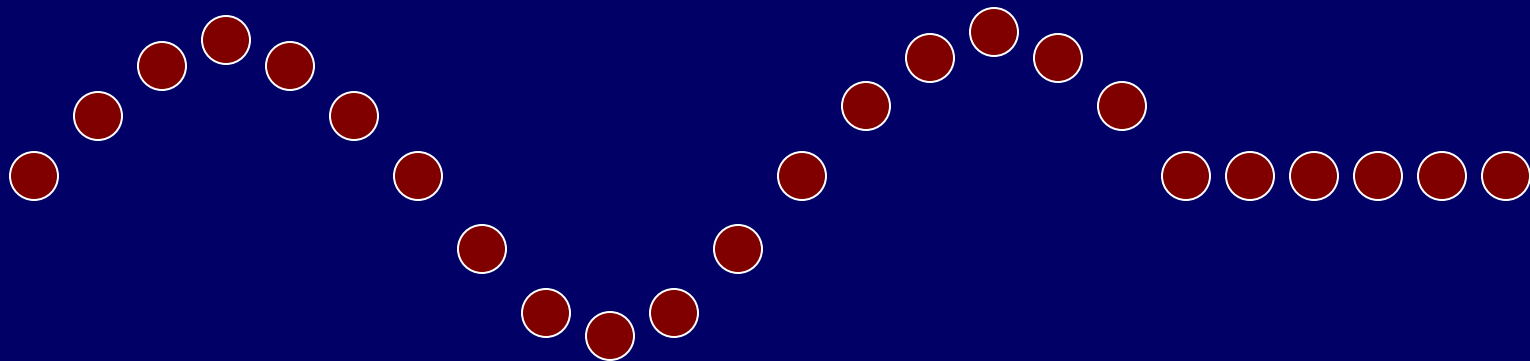
Traveling Wave



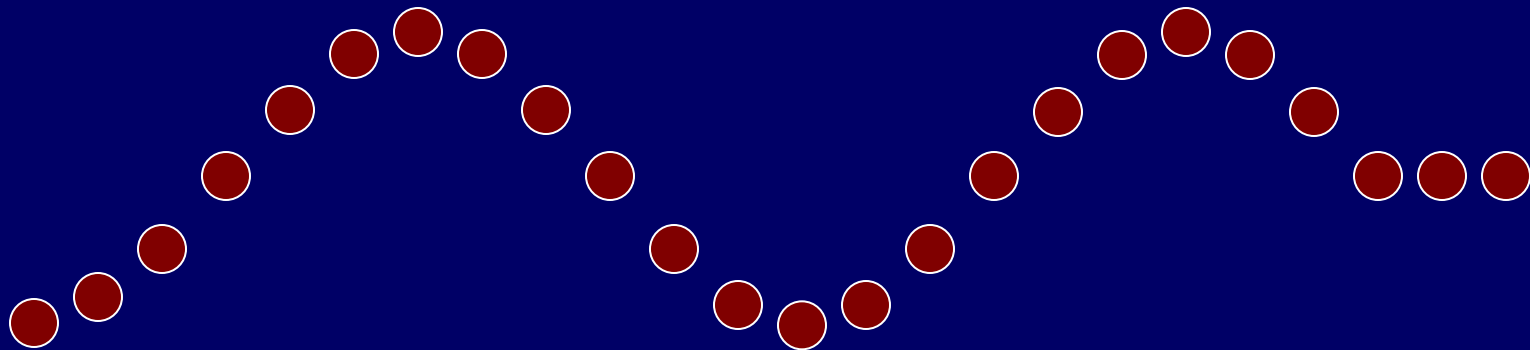
Traveling Wave

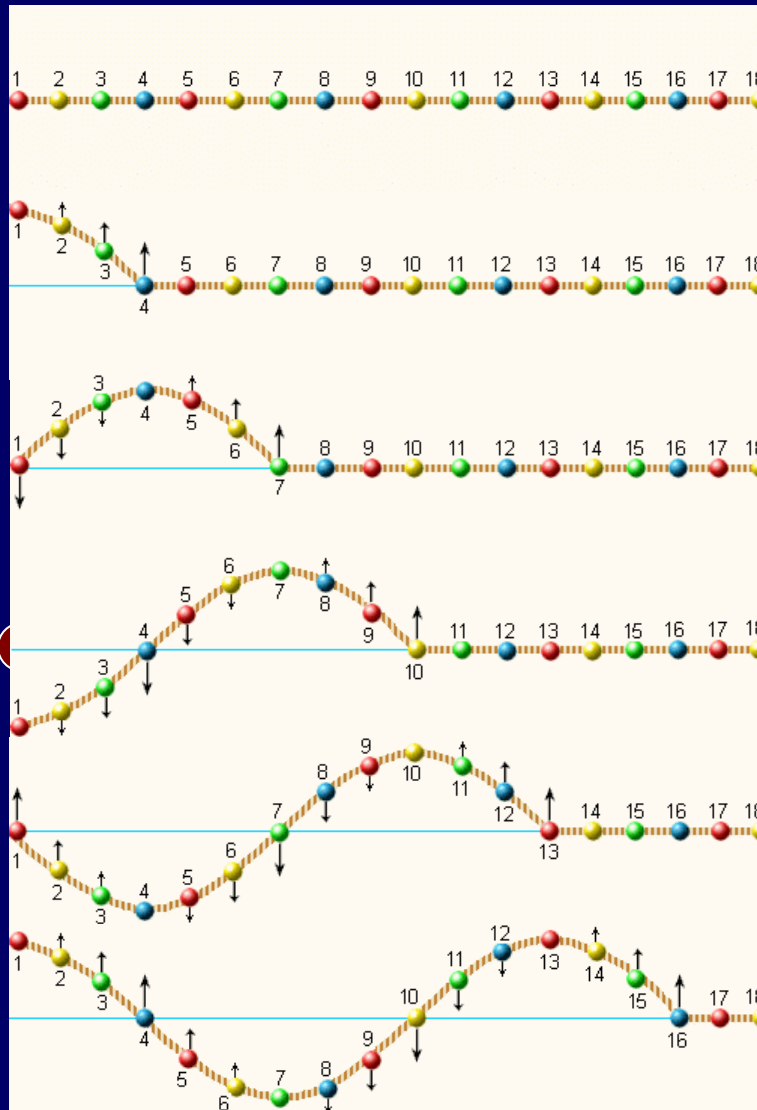


Traveling Wave



Traveling Wave





g Wave

Wave motion

Transfer oscillation

Transfer phase

Not transfer particle



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“Doing the wave”

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Conclusion

1. As the wave passed, the particles of medium do not experience any net displacement with the wave, they only oscillates about their equilibrium positions;
2. Every particle is undergoing the same simple harmonic motion with different initial phase (if neglect the energy dissipation)

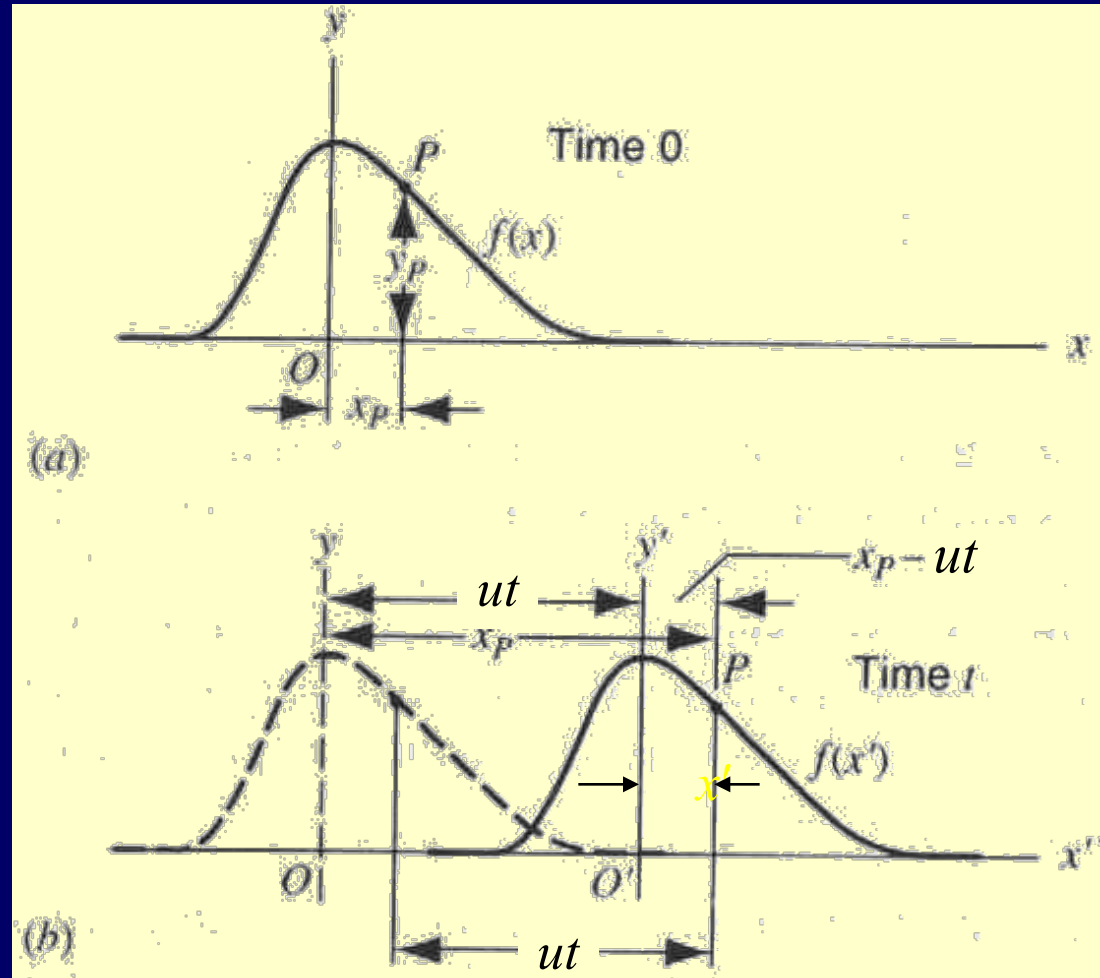
1. Wave function

$$y(x, 0) = f(x)$$

$$y(x, t) = f(x')$$

$$\because x' = x - ut$$

$$\begin{aligned}\because y(x, t) &= f(x') \\ &= f(x - ut)\end{aligned}$$



This is general wave equation. It describes the wave traveling along the positive x direction.

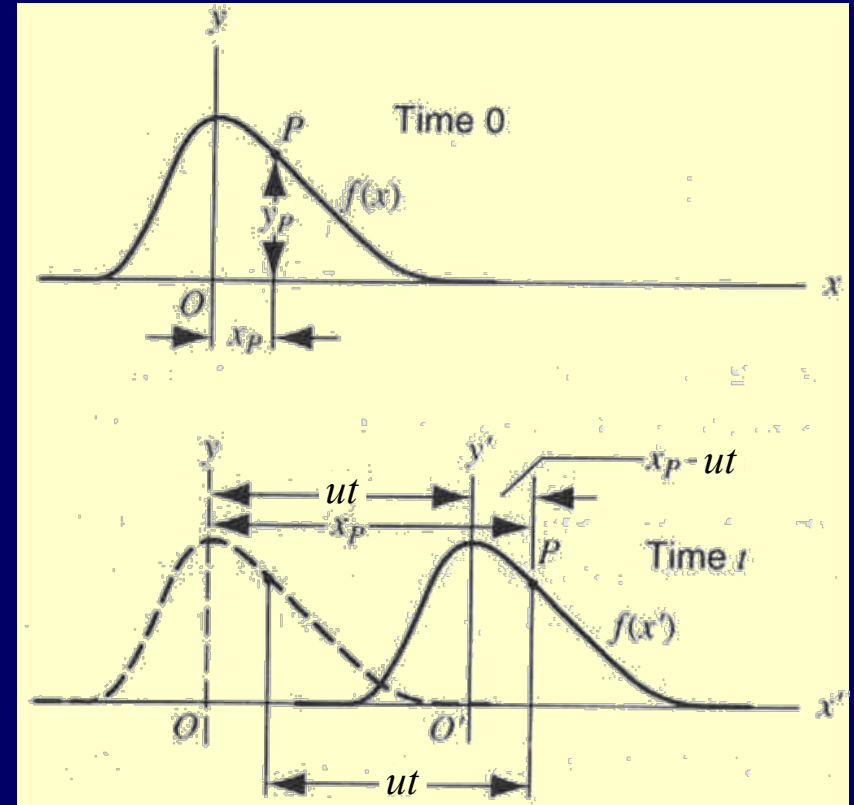
$$y(x, t) = f(x - ut)$$

The velocity of the point P :

$$x - ut = \text{constant}$$

$$dx - udt = 0$$

Phase velocity $u = \frac{dx}{dt}$

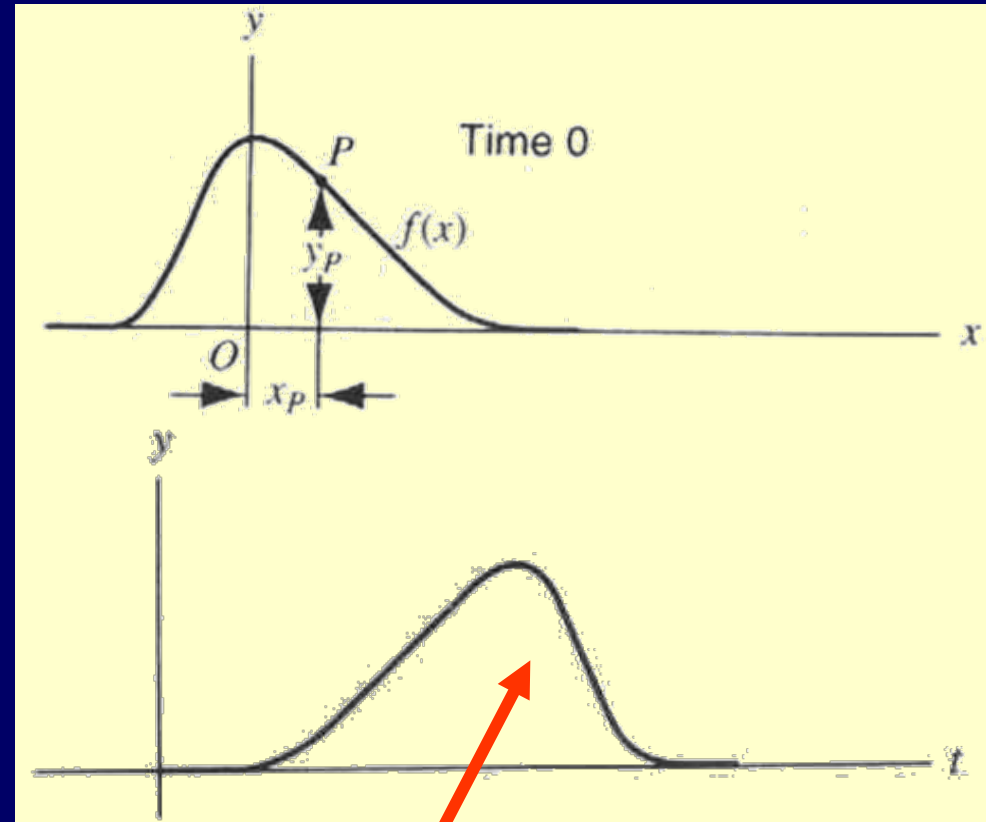


If the wave moves in the negative x direction,

$$y(x, t) = f(x + ut)$$

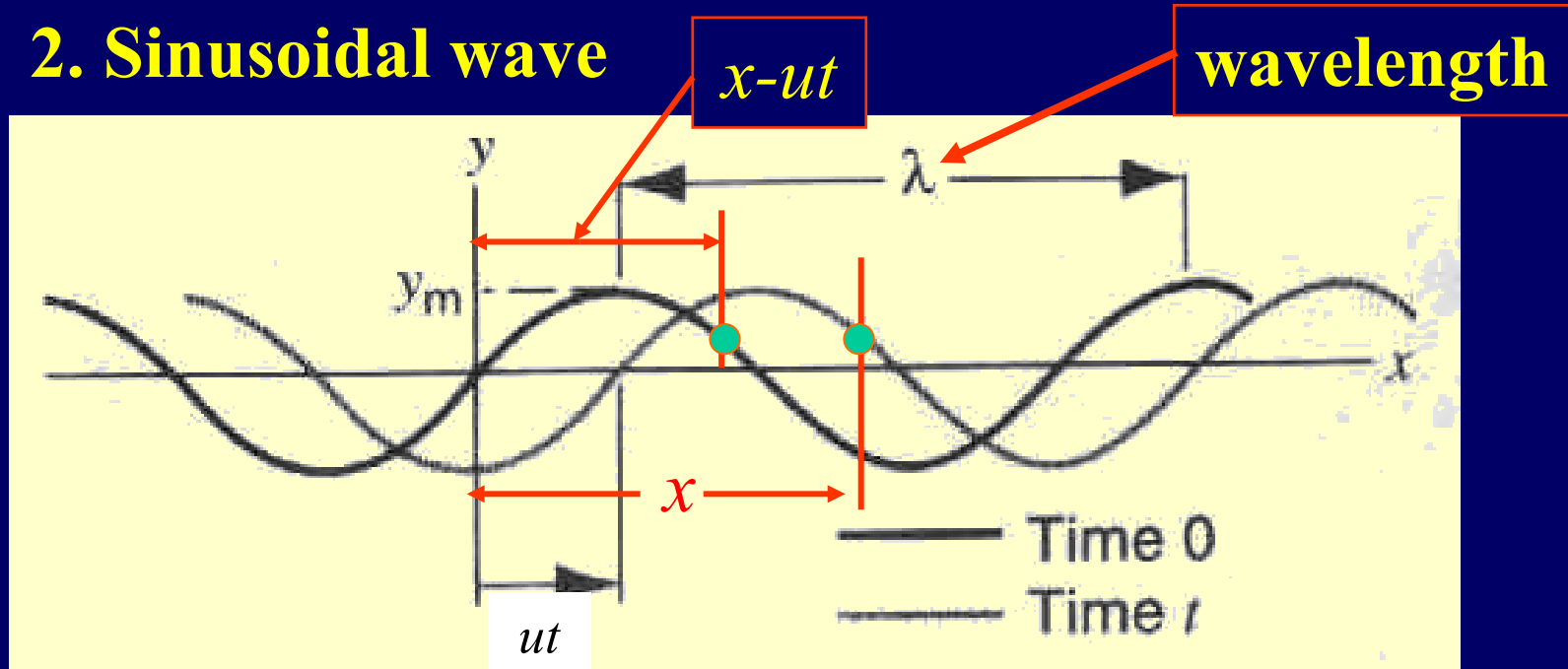
$y=y(x, t_1)$ “snapshot”,
the actual shape of the wave
at time $t_1 = 0$.

$y=y(x_1, t)$ the y
coordinate as a function of
the time at point x_1 .



Note that the waveform!!

2. Sinusoidal wave



$$y(x, 0) = y_m \sin(kx) \quad k = 2\pi / \lambda \quad \text{Angular wave number}$$

波数

$$\therefore y(x, t) = f(x - ut)$$

$$\therefore y(x, t) = y_m \sin[k(x - ut)]$$

$$ku = \omega, \quad u = \omega / k$$

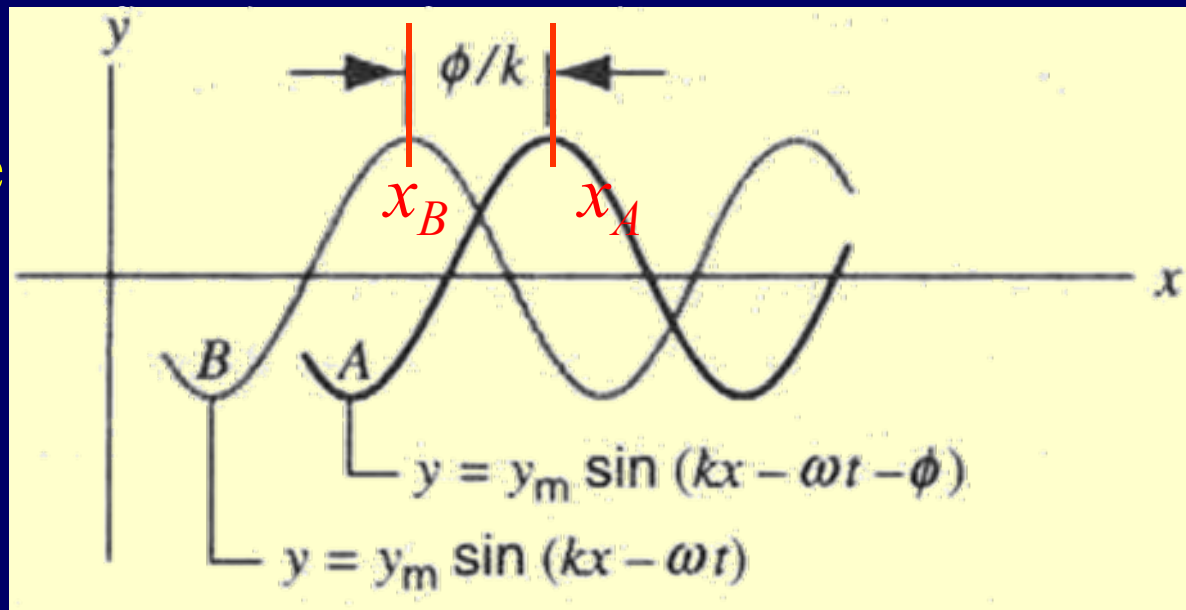
$$u = \lambda / T = f\lambda$$

$$y(x, t) = y_m \sin(kx - \omega t) \text{ or } y = y_m \sin[\omega(t - x / u) + \phi]$$

The phase constant ϕ does not affect the shape of the wave, it moves the wave forward or backward in space or time.

Lead or lag of two phase constants

Two sinusoidal waves traveling in the positive x direction,



$$y_A(x_A, t)$$

$$= y_m \sin(kx_A - \omega t - \phi)$$

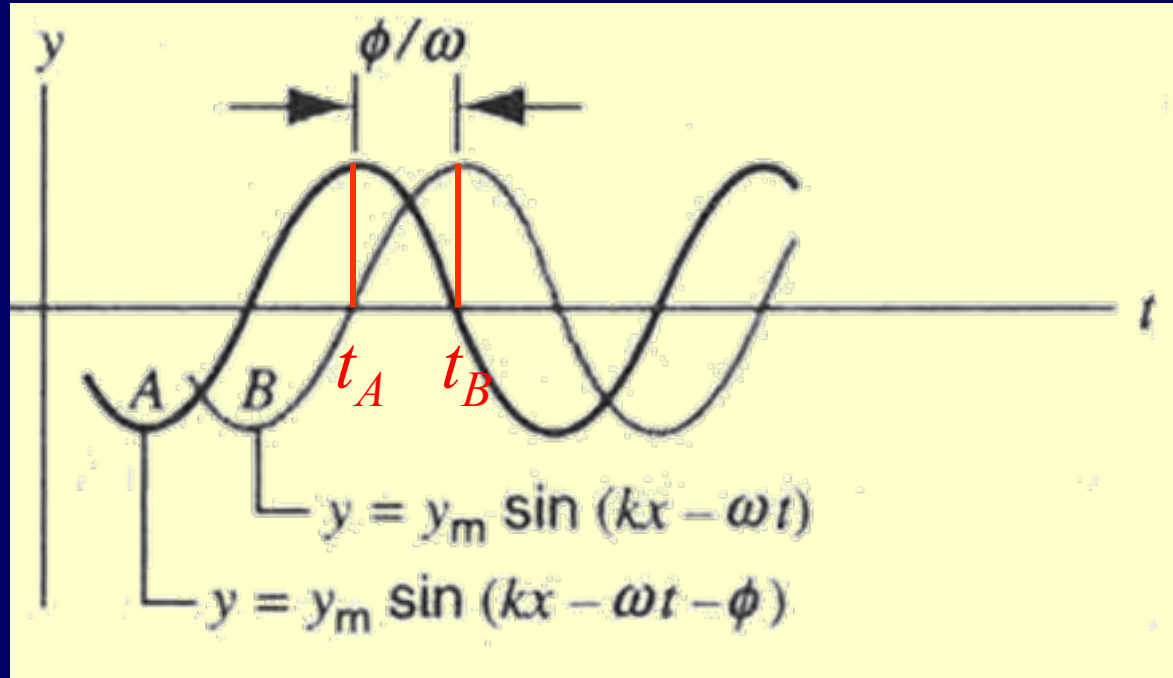
$$= y_m \sin[k(x_A - \phi/k) - \omega t]$$

$$y_B(x_B, t) = y_m \sin(kx_B - \omega t)$$

$$x_A - \frac{\phi}{k} \rightarrow x_B$$

A “**Lead**” (ahead of) B

Two sinusoidal waves traveling in the positive x direction



$$y_A(x, t_A)$$

$$= y_m \sin(kx - \omega t_A - \phi)$$

$$= y_m \sin[kx - \omega(t_A + \phi / \omega)]$$

$$y_B(x, t_B) = y_m \sin(kx - \omega t_B)$$

$$t_A + \frac{\phi}{\omega} \rightarrow t_B$$

A “**Lead**” (ahead of) B

$$y(x, t) = y_m \sin(kx - \omega t - \varphi)$$

If the phase constant is positive, the wave with larger positive phase constant φ is ahead of a wave with smaller positive phase constant. This case is said to “**lead**”. Otherwise it is said to “**Lag**”

If the wave moves in negative direction

$$y(x, t) = y_m \sin(kx + \omega t + \varphi)$$

2. Group speed and dispersion 色散(not dissipation耗散)

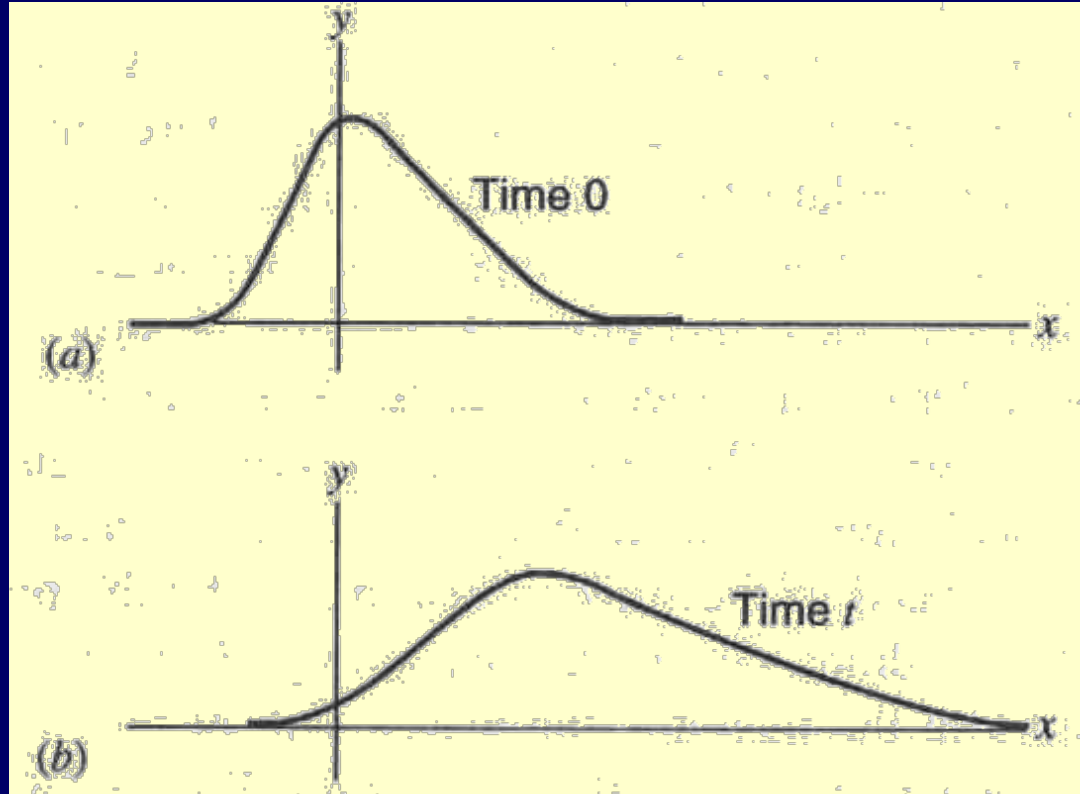
Phase velocity:

$$u_{ph} = \frac{\omega}{k}$$

Group speed:

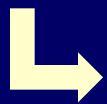
$$u_g = \frac{d\omega}{dk}$$

Electromagnetic wave
propagates in vacuum



$$\omega = ck \longrightarrow u_{ph} = u_g$$

Nondispersive wave



Dispersion relation 色散关系

Example: Electromagnetic wave propagates in plasma

Its dispersion relation: $\omega^2 = \omega_{pe}^2 + c^2 k^2$

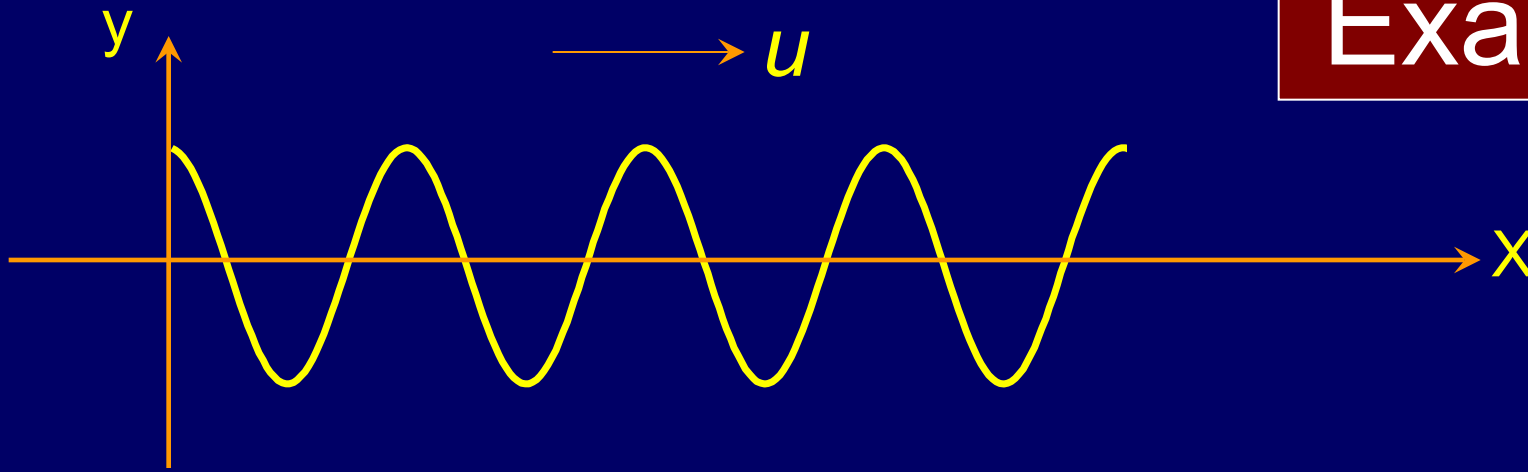
Phase velocity: $u_{ph} = \frac{\omega}{k} = c \sqrt{1 + \frac{\omega_{pe}^2}{c^2 k^2}} > c$

Group speed: $u_g = \frac{d\omega}{dk} = \frac{c}{\sqrt{1 + \frac{\omega_{pe}^2}{c^2 k^2}}} = c \left(1 - \frac{\omega_{pe}^2}{2c^2 k^2}\right) < c$

$$u_{ph} \neq u_g$$

This wave is dispersive

Example



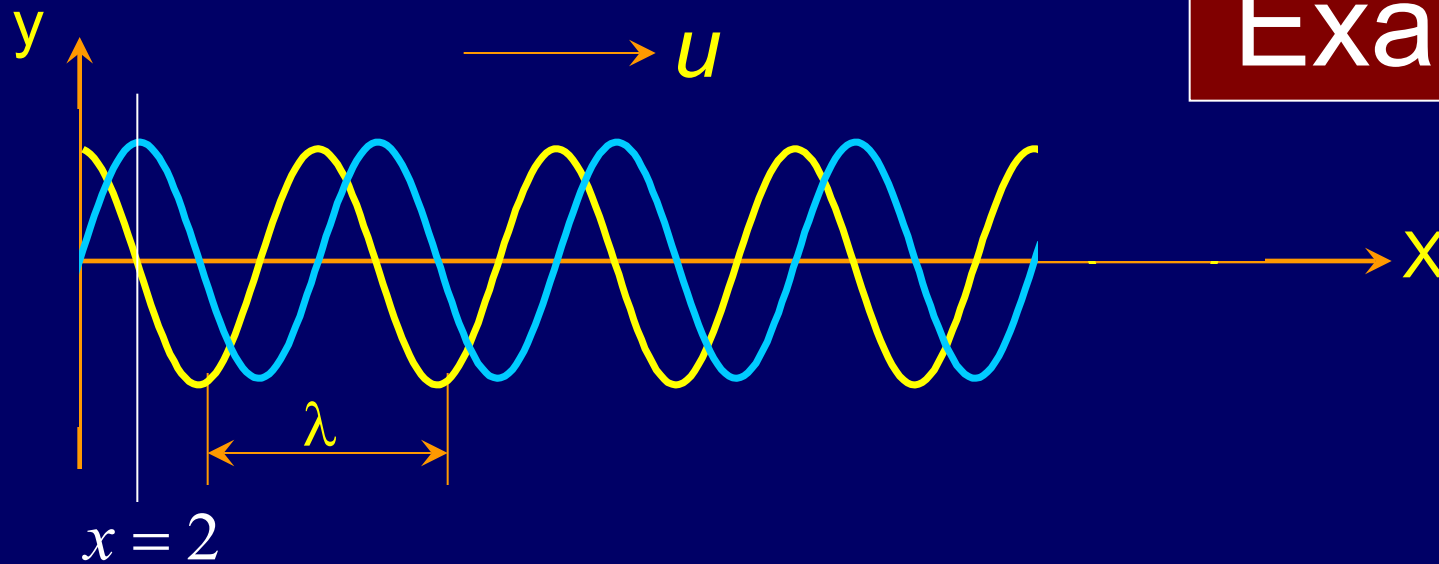
$$\omega = \frac{\pi}{2}, \quad u = 4$$

When $t = 0$, $x = 0$, $y = 2$, it is maximum

What are motion function $k = \omega / u = \pi / 8$

$$y = y_m \sin[(kx - \omega t) + \varphi] \quad y = 2 \sin\left[\frac{\pi}{2}\left(\frac{x}{4} - t\right) + \frac{\pi}{2}\right]$$

Example



When $t = 0$, $x = 0$, $y = 2$, it is maximum $y_m = 2$

After 1 second, $x = 0$, $y = 0$, $T = 4$,

and the particle of $x = 2$ is at maximum $\omega = \frac{2\pi}{T} = \frac{\pi}{2}$

What are $y_m, T, \omega, u, \lambda$

$$u = 2 \quad \lambda = uT = 8$$

Example

$$u = 0.5 \text{ m/s}$$



When $t = 0$

What are motion function

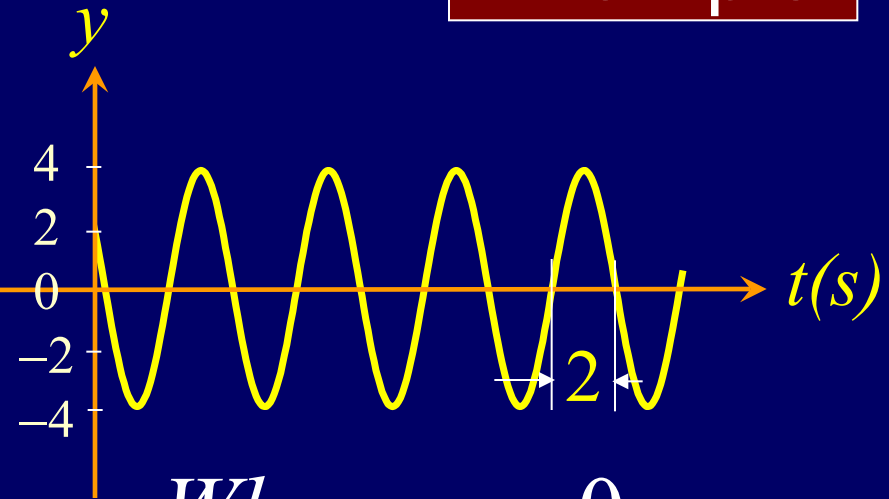
$$y = y_m \sin[(kx - \omega t) + \phi]$$

$$\phi = \frac{5\pi}{6} \quad k = \frac{\pi}{2} \quad \omega = 2\pi \frac{u}{\lambda} = \frac{\pi}{4}$$

$$y = 4 \sin\left[\left(\frac{\pi}{2}x - \frac{\pi}{4}t\right) + \frac{5\pi}{6}\right]$$

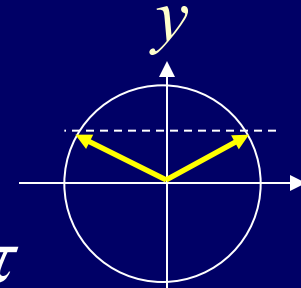
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When $x = 0$

What are motion function

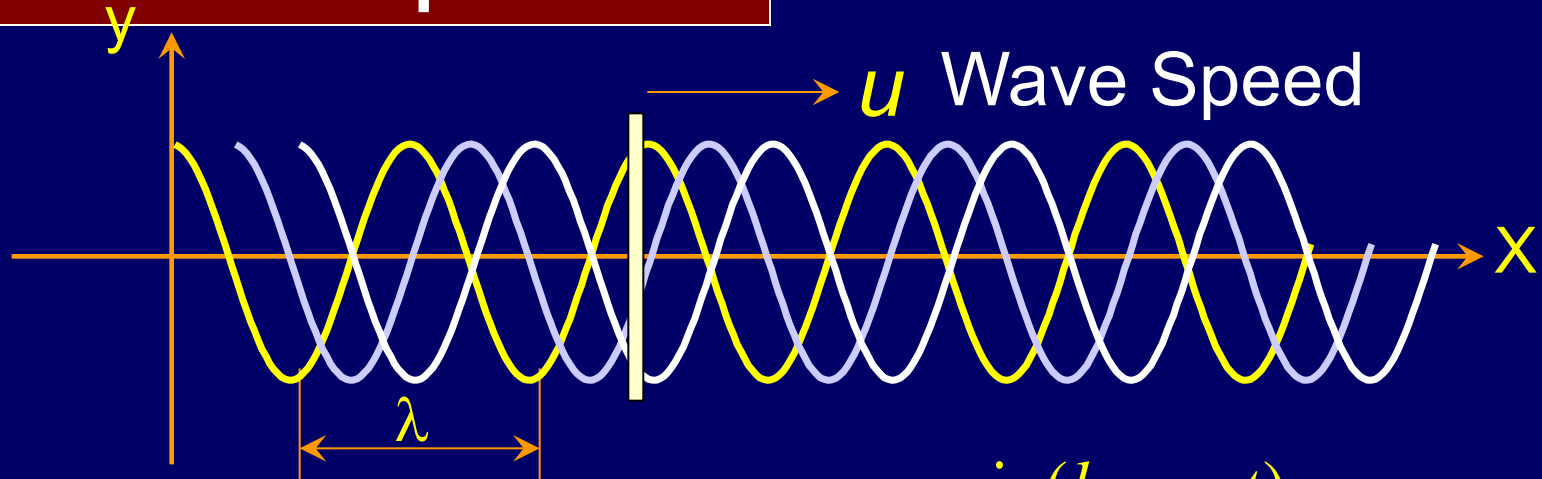


$$\phi = \frac{\pi}{6} \quad T = 4$$

$$\omega = \frac{2\pi}{T} = \frac{\pi}{2} \quad k = \pi$$

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Wave Speed



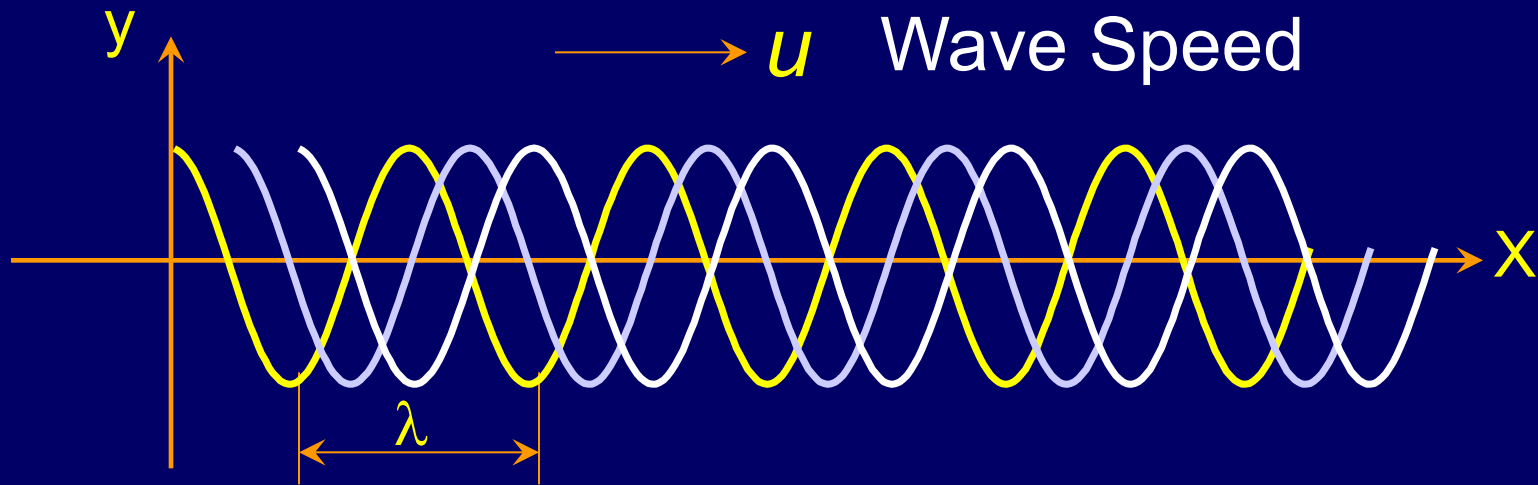
$$y = y_m \sin(kx - \omega t)$$

$$v_y(x, t) = \frac{\partial y}{\partial t} = \frac{\partial}{\partial t} [y_m \sin(kx - \omega t)] = -y_m \omega \cos(kx - \omega t)$$

$$a_y(x, t) = \frac{\partial^2 y}{\partial t^2} = \frac{\partial v_y}{\partial t} = -y_m \omega^2 \sin(kx - \omega t) = -\omega^2 y$$

$$\frac{\partial^2 y}{\partial t^2} + \omega^2 y = 0$$

Wave Speed



$$y = y_m \sin[(kx - \omega t) + \phi]$$

$$u \neq \frac{dy}{dt} = -\omega y_m \cos[(kx - \omega t) + \phi]$$

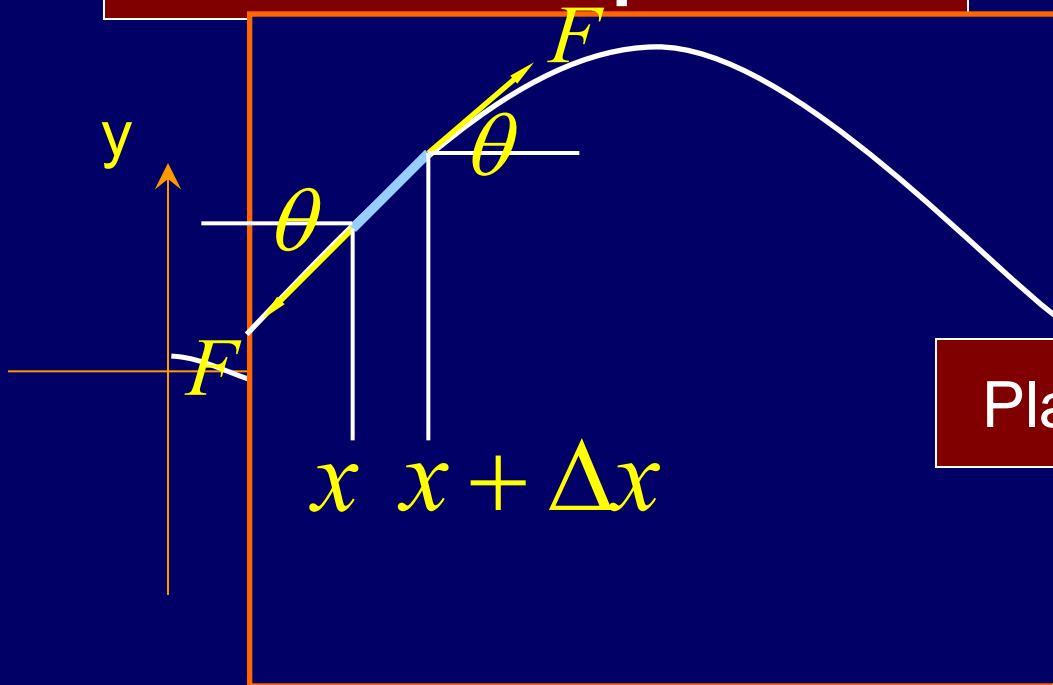
$$u = ?$$

Velocity of oscillation

Wave Speed

$$u = \sqrt{\frac{F}{\mu}}$$

$$u^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}$$



$$y = y_m \sin[(kx - \omega t) + \varphi]$$

Plane Simple Harmonic Wave

$$\Sigma F = F \sin \theta \Big|_{x+\Delta x} - F \sin \theta \Big|_x = F \frac{\Delta y}{\Delta x} \Big|_{x+\Delta x} - F \frac{\Delta y}{\Delta x} \Big|_x = \mu \Delta x \frac{d^2 y}{dt^2}$$

$$\frac{F}{\mu} \frac{\frac{\Delta y}{\Delta x} \Big|_{x+\Delta x} - \frac{\Delta y}{\Delta x} \Big|_x}{\Delta x} = \frac{d^2 y}{dt^2}$$

$$\Delta x \rightarrow 0$$

$$\frac{F}{\mu} \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2}$$

10.1 一维波动方程的通解

本节我们一般性地求解波动方程

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad (10.1.1)$$

在 5.4.2 节例 2 中, 我们将方程 (10.1.1) 化成了标准形式 (5.4.57)。其实该过程也可以按下面的方法直接进行, 令

$$\xi = x + at, \quad \eta = x - at \quad (10.1.2)$$

利用复合函数微分法则, 得到

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \\ \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial \xi} \left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \right) \frac{\partial \xi}{\partial x} + \frac{\partial}{\partial \eta} \left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \right) \frac{\partial \eta}{\partial x} \\ &= \frac{\partial}{\partial \xi} \left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \right) + \frac{\partial}{\partial \eta} \left(\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta} \right) \\ &= \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2} \end{aligned} \quad (10.1.3a)$$

同理可得

$$\frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial \xi^2} - 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2} \right) \quad (10.1.3b)$$

将式 (10.1.3) 代入式 (10.1.1) 得到

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0 \quad (10.1.4)$$

现在求方程 (10.1.4) 的通解, 首先对 η 积分得到

$$\frac{\partial u}{\partial \xi} = C_1 \quad (10.1.5)$$

其中, C_1 是对 η 积分而出现的积分常数, 因此 C_1 不含 η 。但 C_1 必须包含 ξ [否则方程 (10.1.5) 仅给出 $u(\xi, \eta) \propto \xi$ 形式的结果], 于是式 (10.1.5) 可以写为

$$\frac{\partial u}{\partial \xi} = f(\xi) \quad (10.1.6)$$

其中, $f(\xi)$ 是 ξ 的任意可微函数。再将式 (10.1.6) 对 ξ 积分, 得到

$$u(\xi, \eta) = \int f(\xi) d\xi + C_2 \quad (10.1.7)$$

其中, C_2 是对 ξ 积分而出现的积分常数, 因此 C_2 不含 ξ 。但 C_2 必须包含 η [否则方程 (10.1.7) 给出的结果只包含 $u(\xi, \eta)$ 对 ξ 的依赖性, 与 η 无关]。这样式 (10.1.7) 可以写为

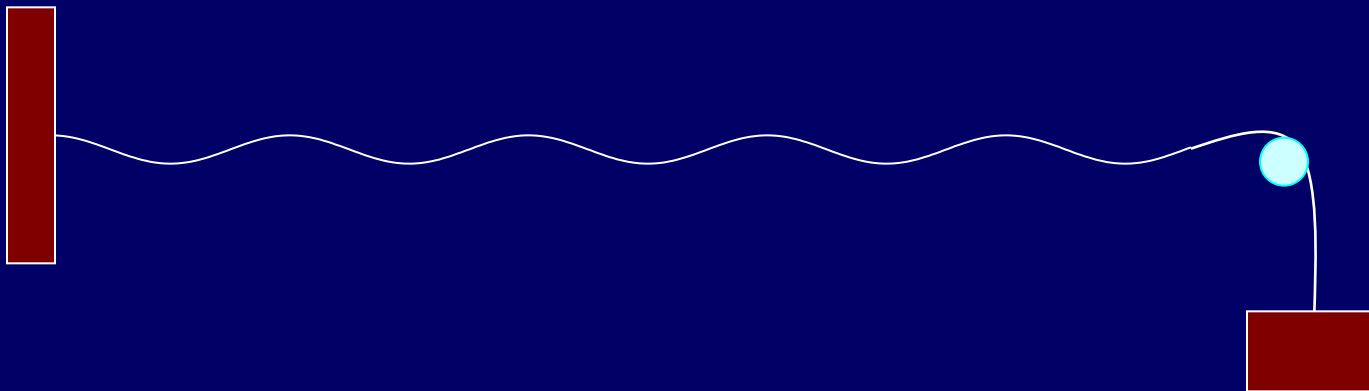
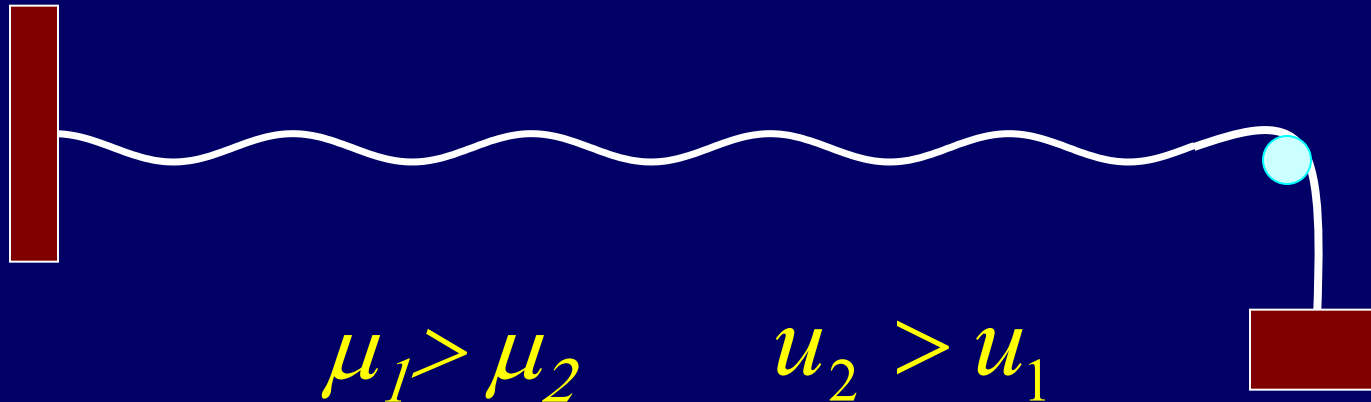
$$u(\xi, \eta) = f_1(\xi) + f_2(\eta) \quad (10.1.8)$$

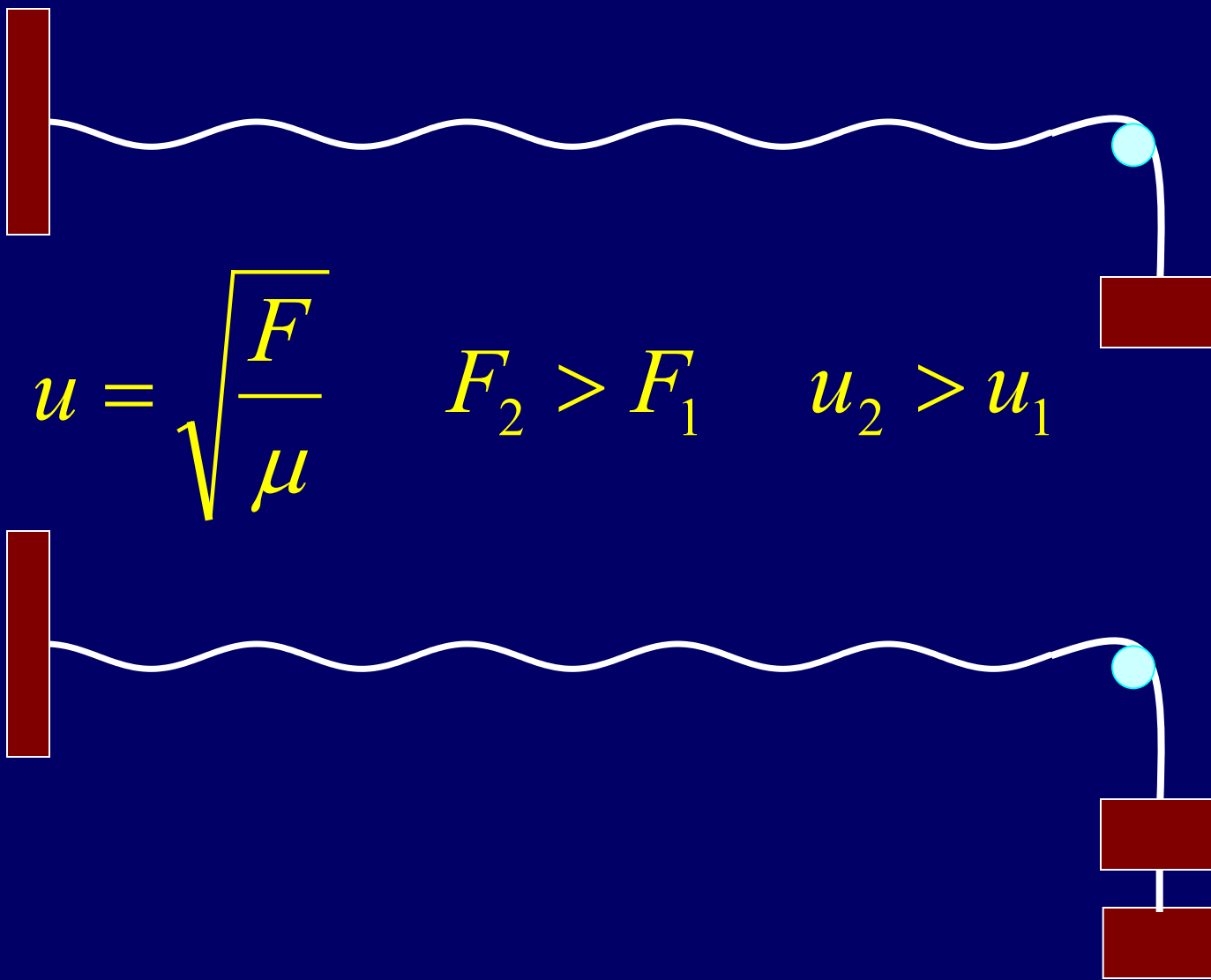
即

$$u(x, t) = f_1(x + at) + f_2(x - at) \quad (10.1.9)$$

这就是方程 (10.1.1) 的解, 其中, f_1, f_2 都是任意二次连续可微函数。我们看到, 式 (10.1.9) 的解满足任意二次连续可微函数, 因此它是波动方程 (10.1.1) 的通解 (包含有两

$$u^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2} \quad u = \sqrt{\frac{F}{\mu}}$$





The energy of mechanical wave
Interference of mechanical wave
Standing wave
Reflection of mechanical wave
Resonance of mechanical wave

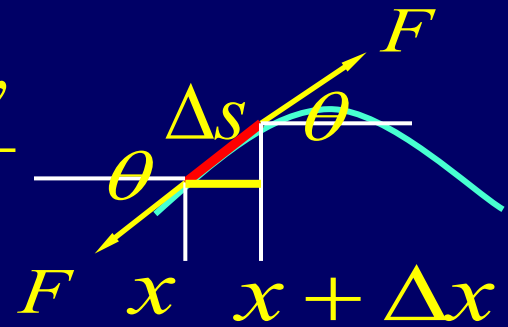
Wave Energy

$$\frac{F}{\mu} \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} \quad u = \sqrt{\frac{F}{\mu}}$$

$$\Delta K = \frac{1}{2} \Delta m v^2$$

$$\Delta K = \frac{1}{2} \mu \Delta x \left(\frac{dy}{dt} \right)^2$$

$$u^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}$$



Kinetic Energy

$$\Delta K = \frac{1}{2} \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$$

$$s - \Delta x \approx \frac{1}{2} \left(\frac{\partial y}{\partial x} \right)^2 \Delta x$$

$$Fk^2 = \mu u^2 k^2 = \mu \hat{\omega}^2$$

Potential Energy

$$y = y_m \sin[(kx - \omega t) + \varphi]$$

$$\Delta U = \frac{1}{2} \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$$

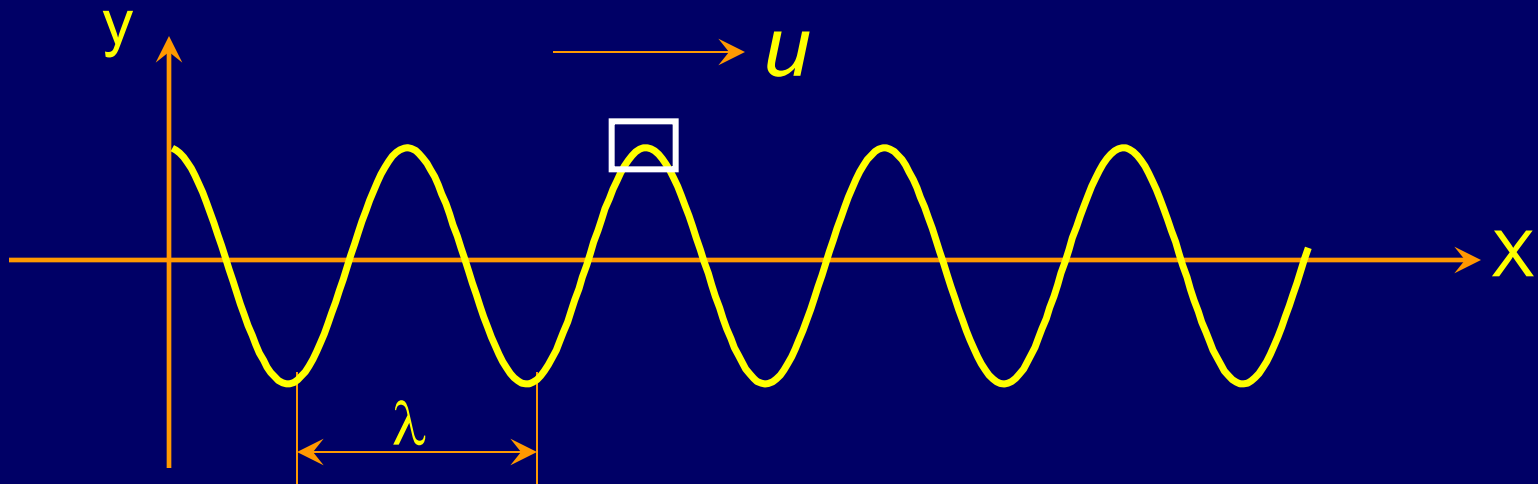
$$\frac{dy}{dx} = y_m k \cos[(kx - \omega t) + \varphi]$$

$$\Delta K = \Delta U$$

$$\frac{dy}{dt} = -y_m \omega \cos[(kx - \omega t) + \varphi]$$

$$\Delta K + \Delta U = \mu \Delta x y_m^2 \omega^2 \cos^2[\omega(t - \frac{x}{u}) + \varphi]$$

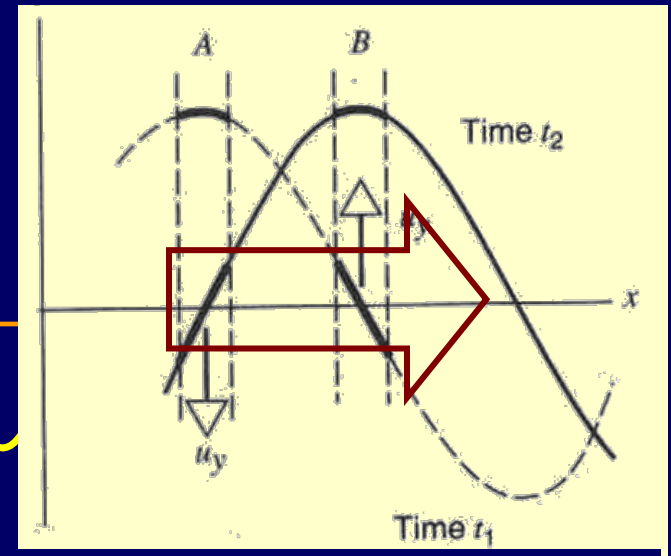
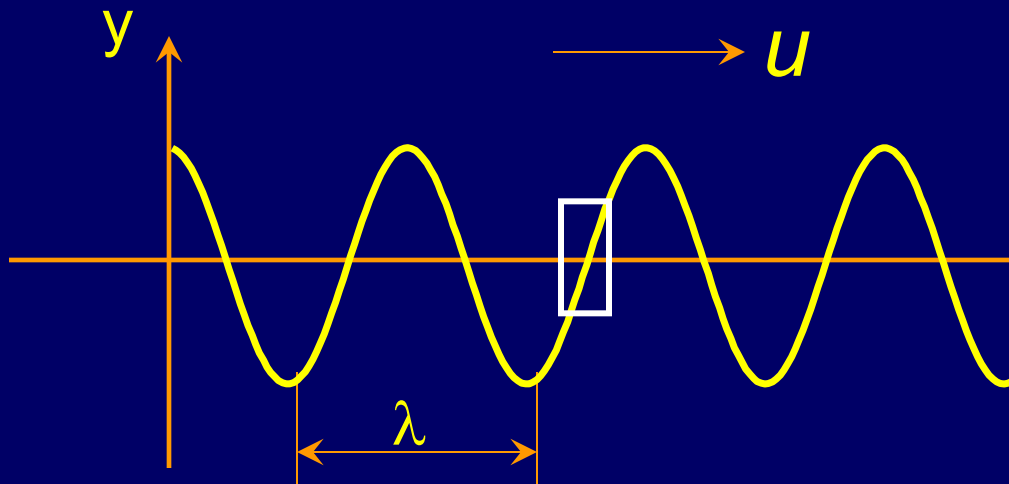
$$\Delta K + \Delta U = \Delta E \neq \text{const.}$$



$$\Delta K = \frac{1}{2} \mu \Delta x \left(\frac{dy}{dt} \right)^2 \quad v \rightarrow 0, \quad \Delta K \rightarrow 0$$

$$\Delta U = \frac{F}{2} \left(\frac{\partial y}{\partial x} \right)^2 \Delta x \quad \frac{\Delta y}{\Delta x} \rightarrow 0, \quad \Delta U \rightarrow 0$$

$$t = t_1, \quad K, U(A) \rightarrow \min \quad K, U(B) \rightarrow \max$$



$$t = t_2, \quad K, U(A) \rightarrow \max$$

Transfer oscillation

$$\Delta K = \frac{1}{2} \mu \Delta x \left(\frac{dy}{dt} \right)^2 \quad v \rightarrow v_{\max},$$

Transfer phase

Transfer energy

$$\Delta U = \frac{F}{2} \left(\frac{\partial y}{\partial x} \right)^2 \Delta x \quad \frac{\Delta y}{\Delta x} \rightarrow \max, \quad U \rightarrow U_{\max}$$

Plane Simple Harmonic Wave

Wave Function

$$u^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}$$

$$y = y_m \sin[(kx - \omega t) + \varphi]$$

$$\omega = \frac{2\pi}{T} \quad \lambda = uT \quad k = \frac{\omega}{u}$$

Transfer oscillation

Transfer phase Transfer energy

Wave Speed



$$u = \sqrt{\frac{F}{\mu}} \quad \text{string}$$

Wave Energy

$$\Delta K = \frac{1}{2} \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$$

$$E = \Delta K + \Delta U = \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$$

$$\Delta U = \frac{1}{2} \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$$

Wave Energy $E = \Delta K + \Delta U = \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi]$

Average of energy

$$\frac{dy}{dx} = y_m k \cos[(kx - \omega t) + \varphi]$$

$$\overline{\Delta E} = \frac{\int_0^T \Delta E(t) dt}{T} = \int_0^T \mu \Delta x y_m^2 \omega^2 \cos^2[(kx - \omega t) + \varphi] dt / T = \frac{1}{2} \Delta x \mu y_m^2 \omega^2$$

where $\frac{1}{T} \int_0^T \cos^2 \theta d\theta = \frac{1}{2}$ $\frac{dy}{dt} = -y_m \omega \cos[(kx - \omega t) + \varphi]$

Power

$$P = \langle \vec{F} \cdot \vec{v} \rangle = \langle F v \sin \theta \rangle = \left\langle F \frac{\Delta y}{\Delta x} \frac{\Delta y}{\Delta t} \right\rangle = \frac{1}{2} \mu y_m^2 \omega^2 u$$

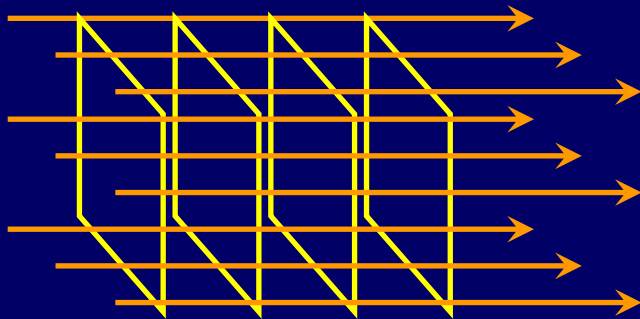
Intensity

$$I = \frac{P}{A}$$

$$\bar{e} = \frac{\bar{E}}{V}$$

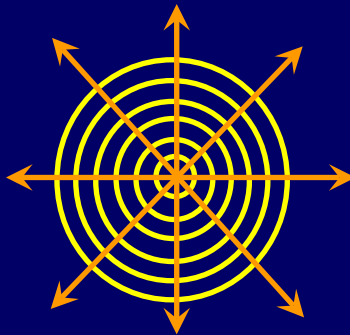
Energy Density

Wave Energy



$$I = I_0$$

plane wave



$$I = \frac{I_0}{4\pi r^2}$$

Spherical wave

Wave superposition, interference, and boundary condition

□ The principle of superposition

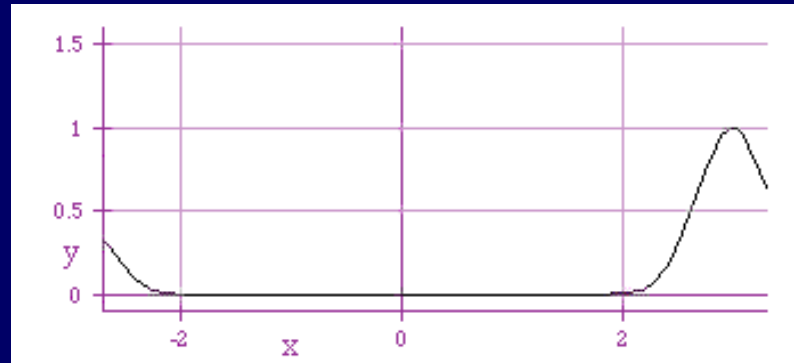
- When two waves overlap, the actual displacement of any point at any time is obtained by adding the displacement the point would have if only the first wave were present and the displacement it would have if only the second wave were present:

$$y(x, t) = y_1(x, t) + y_2(x, t)$$

Wave superposition, interference, and boundary condition (cont'd)

□ Interference

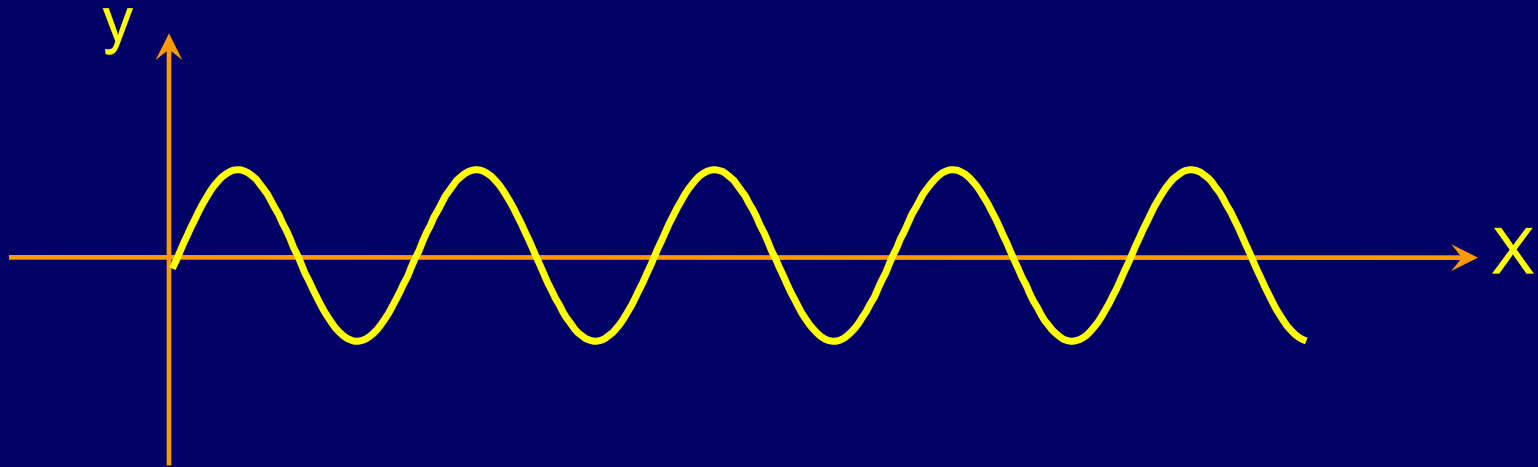
- Constructive interference (positive-positive or negative-negative)



- Destructive interference (positive-negative)

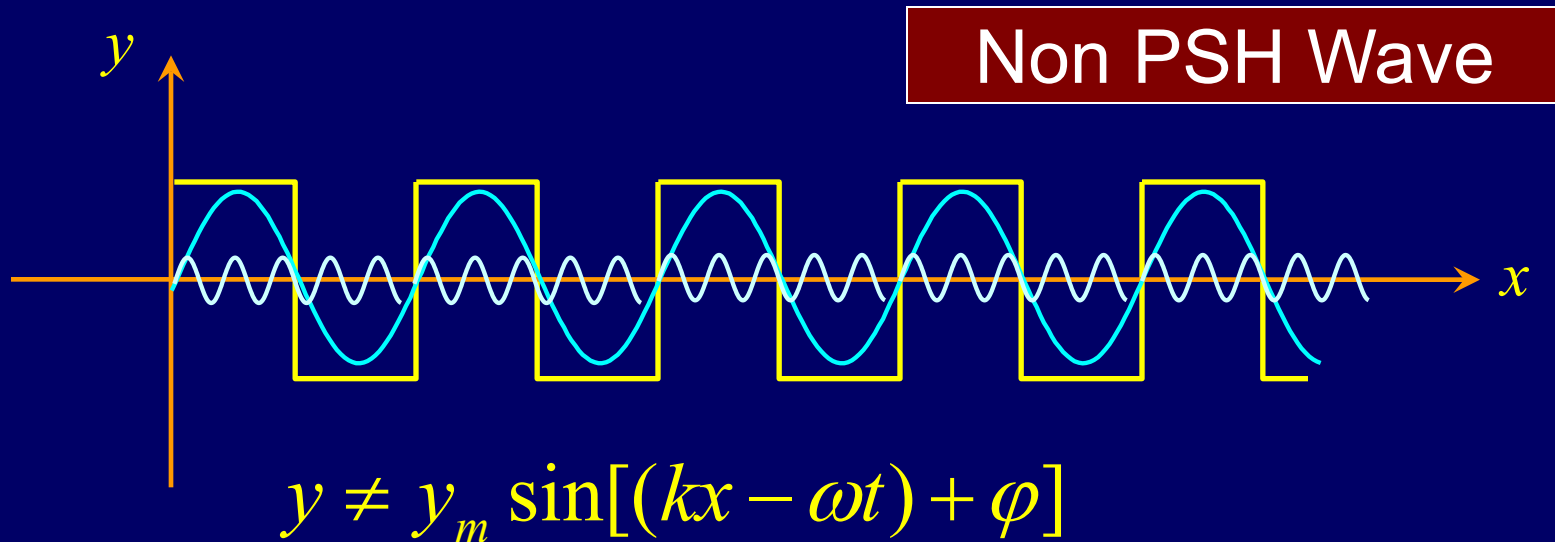


Plane Simple Harmonic Wave



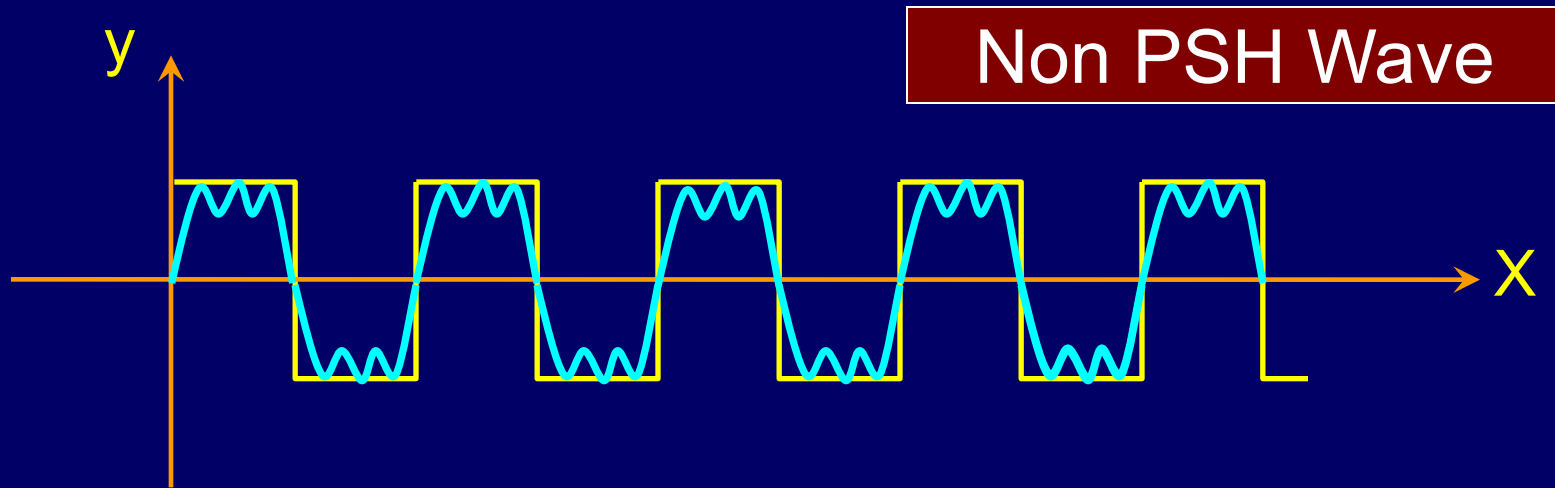
$$y = y_m \sin[(kx - \omega t) + \varphi]$$

Plane Simple Harmonic Wave



$$y = \sum A_n \sin[(k_n x - \omega_n t) + \varphi]$$

Plane Simple Harmonic Wave



$$y \neq y_m \sin[(kx - \omega t) + \varphi]$$

$$y = \sum A_n \sin[(k_n x - \omega_n t) + \varphi]$$

□ Fourier's theorem and frequency spectrum

- Fourier's theorem:

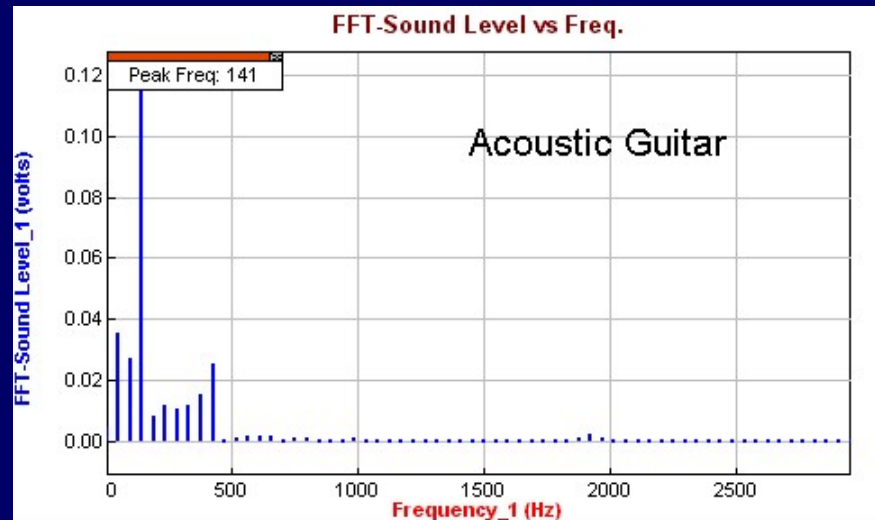
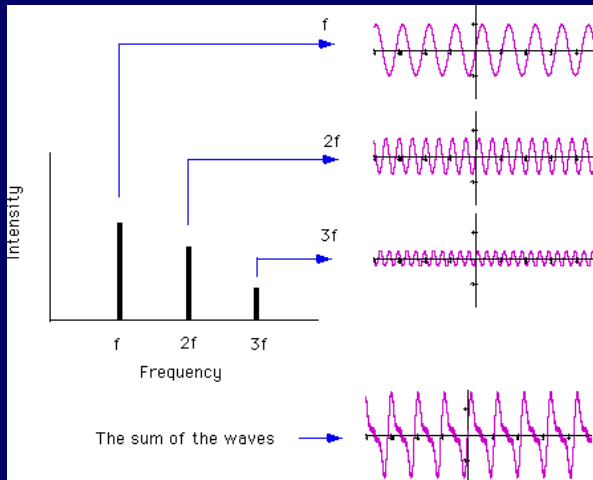
Any periodic function of period T can be written as

$$y(t) = \sum_n [A_n \sin(2\pi f_n t) + B_n \cos(2\pi f_n t)]$$

fundamental freq.

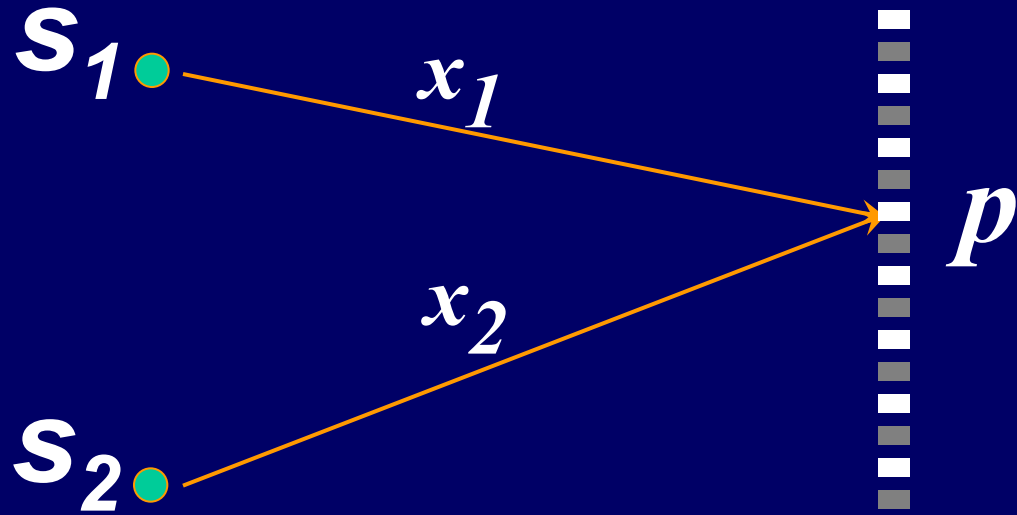
where $f_1 = 1/T$, $f_n = n f_1$ ($n = 1, 2, 3, \dots$)

- Implication of Fourier's theorem:



interference

At p point



$$y_1 = y_m \sin \omega(t - \frac{x_1}{u})$$

$$y_2 = y_m \sin \omega(t - \frac{x_2}{u})$$

if $y_{m1} = y_{m2} \quad \phi_1 = \phi_2 = 0$

$$y = y_1 + y_2 = y_m \sin \omega(t - \frac{x_1}{u}) + y_m \sin \omega(t - \frac{x_2}{u})$$

$$y = 2y_m \cos \pi \frac{x_1 - x_2}{\lambda} \sin (\omega t - \omega \frac{x_1 + x_2}{2u})$$

$$\begin{cases} x_1 - x_2 = n \lambda & \text{max} \\ x_1 - x_2 = (n + 1/2) \lambda & \text{min} \end{cases}$$

Constructive interference
(positive-positive or negative-negative)

Destructive interference
(positive-negative)

Wave interference, boundary condition, and superposition (cont'd)

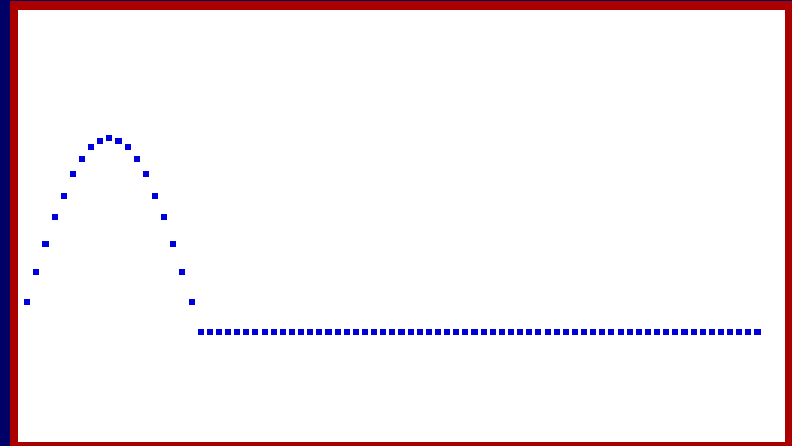
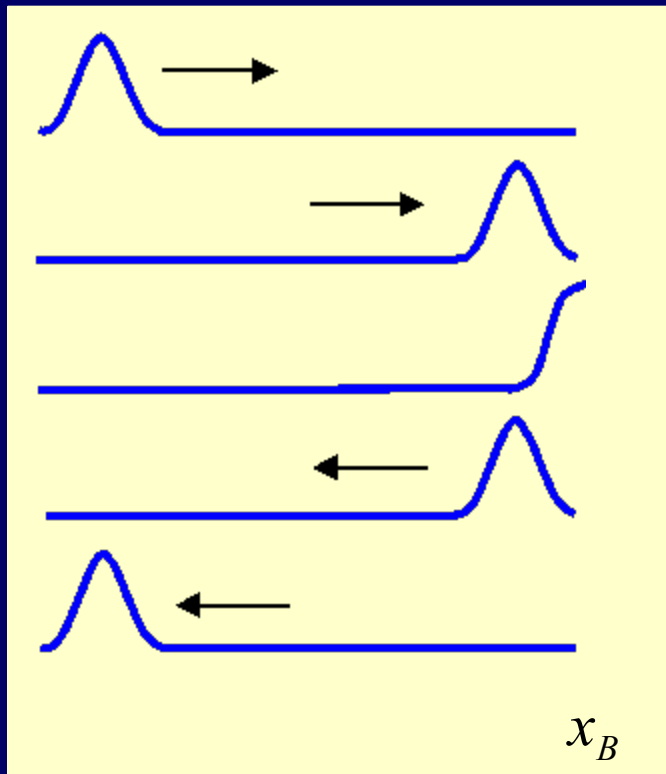
□ Reflection

• Free end (Soft)

incident wave reflected wave

$$y(x,t) = A \sin(kx - \omega t) + B \sin(kx + \omega t + \varphi) ?$$

For $x < x_B$



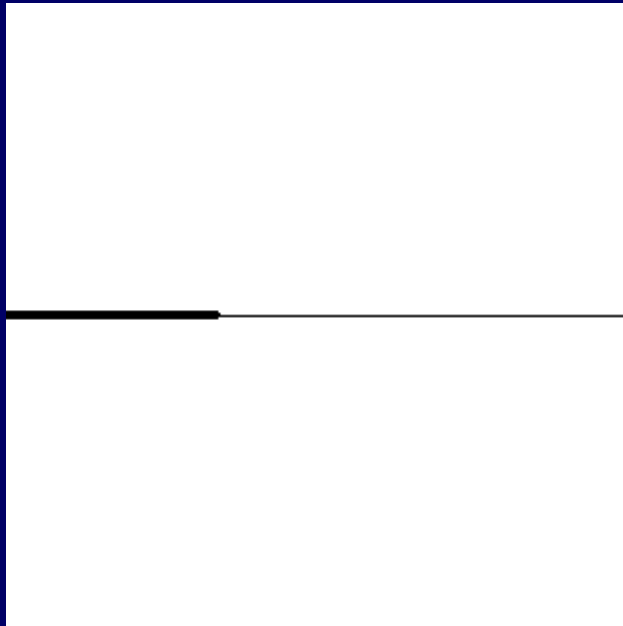
At $x = x_B$ $(\partial y / \partial x)_{x=x_B} = 0 \rightarrow B = +A$

Vertical component of the force
at the boundary is zero.

Wave interference, boundary condition, and superposition (cont'd)

□ Reflection (cont'd)

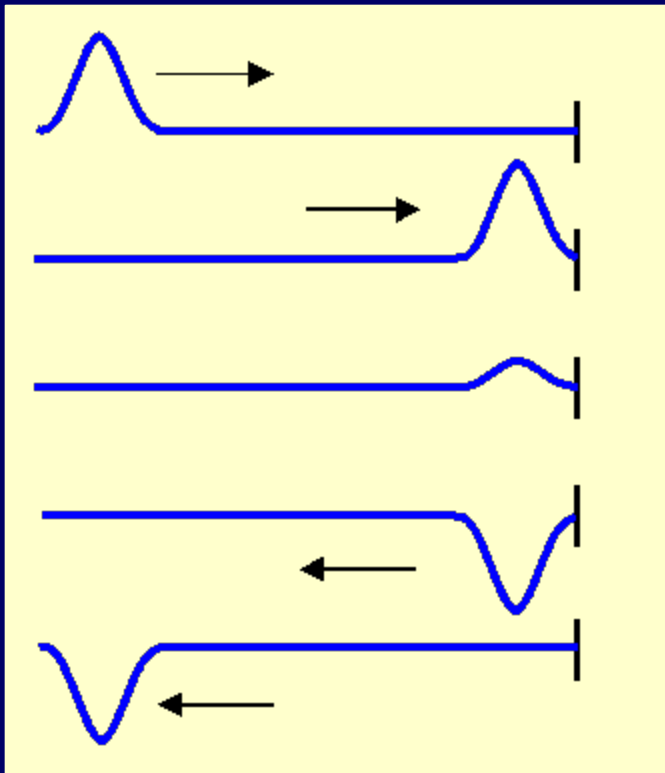
- At high/low density



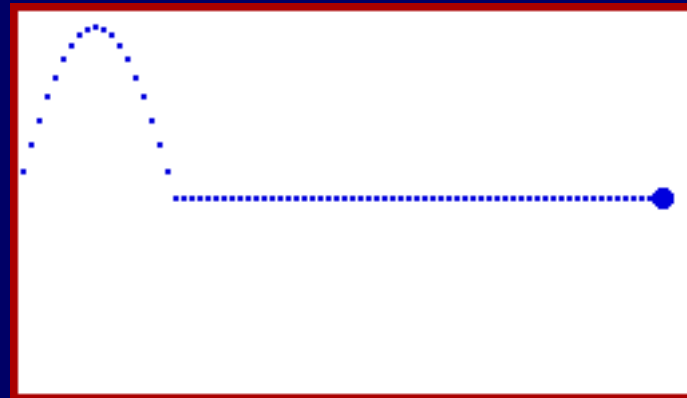
Wave interference, boundary condition, and superposition (cont'd)

□ Reflection (cont'd) $y(x, t) = A \sin(kx - \omega t) + B \sin(kx + \omega t + \overset{?}{\varphi})$

- Fixed end (Hard)



For $x < x_B$



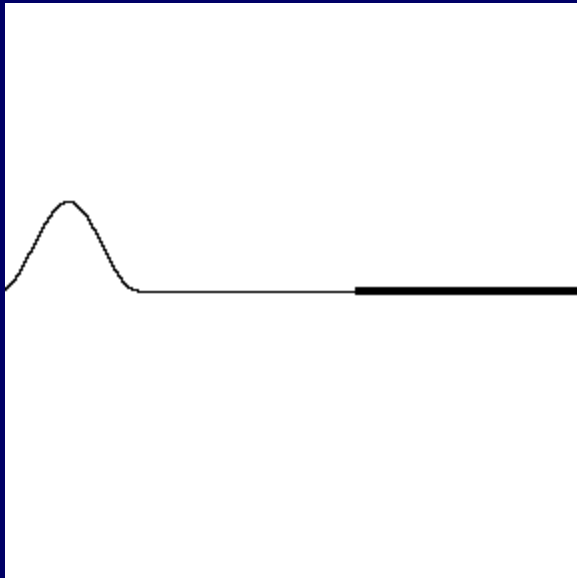
At $x = x_B$ $y(x, t)_{x=x_B} = 0 \rightarrow B = -A$

Displacement at the boundary is zero.

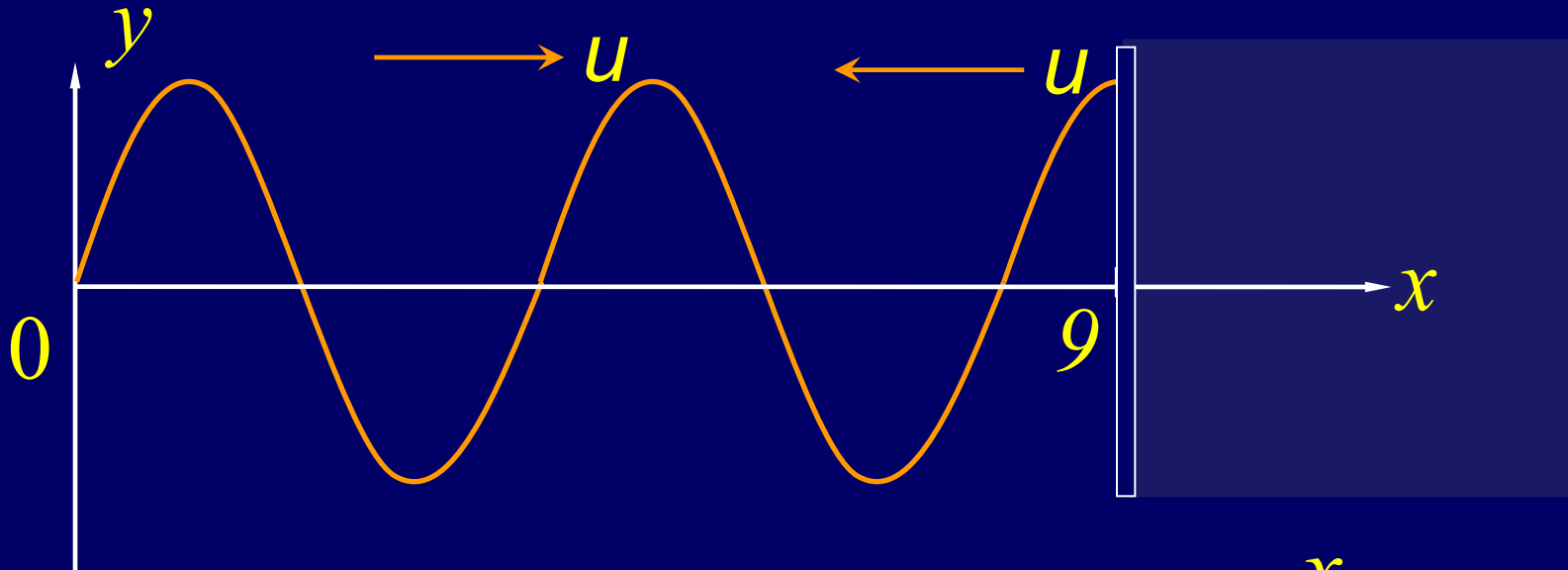
Wave interference, boundary condition, and superposition (cont'd)

□ Reflection (cont'd)

- At low/high density



Reflection



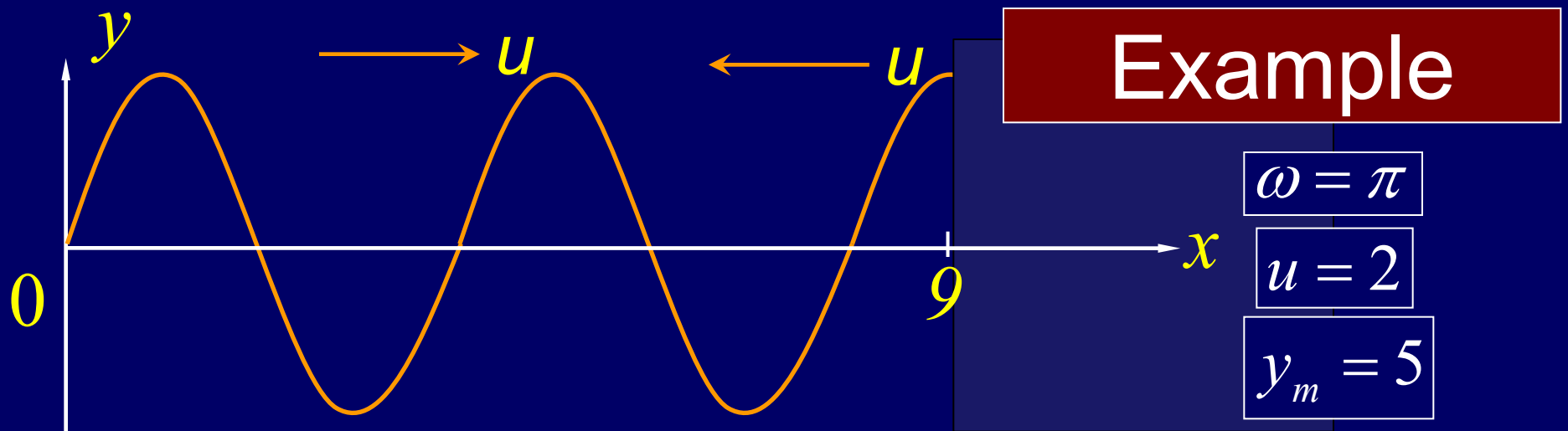
$$\omega = \pi \quad y_m = 5$$

$$k = \pi / 2$$

$$u = 2$$

$$y = y_m \sin\left[\omega\left(t - \frac{x}{u}\right) + \phi\right]$$

$$y = y_m \sin\left[\omega\left(t + \frac{x}{u}\right) + \phi'\right] \quad \phi' = ?$$



$$y = 5 \sin \pi \left(t - \frac{x}{2} \right) \quad y|_{x=9} = 5 \sin \pi \left(t - \frac{9}{2} \right) = 5 \sin \left(\pi t - \frac{\pi}{2} \right)$$

Reflection Wave $y = 5 \sin \left[\pi \left(t + \frac{x}{2} \right) + \phi' \right]$

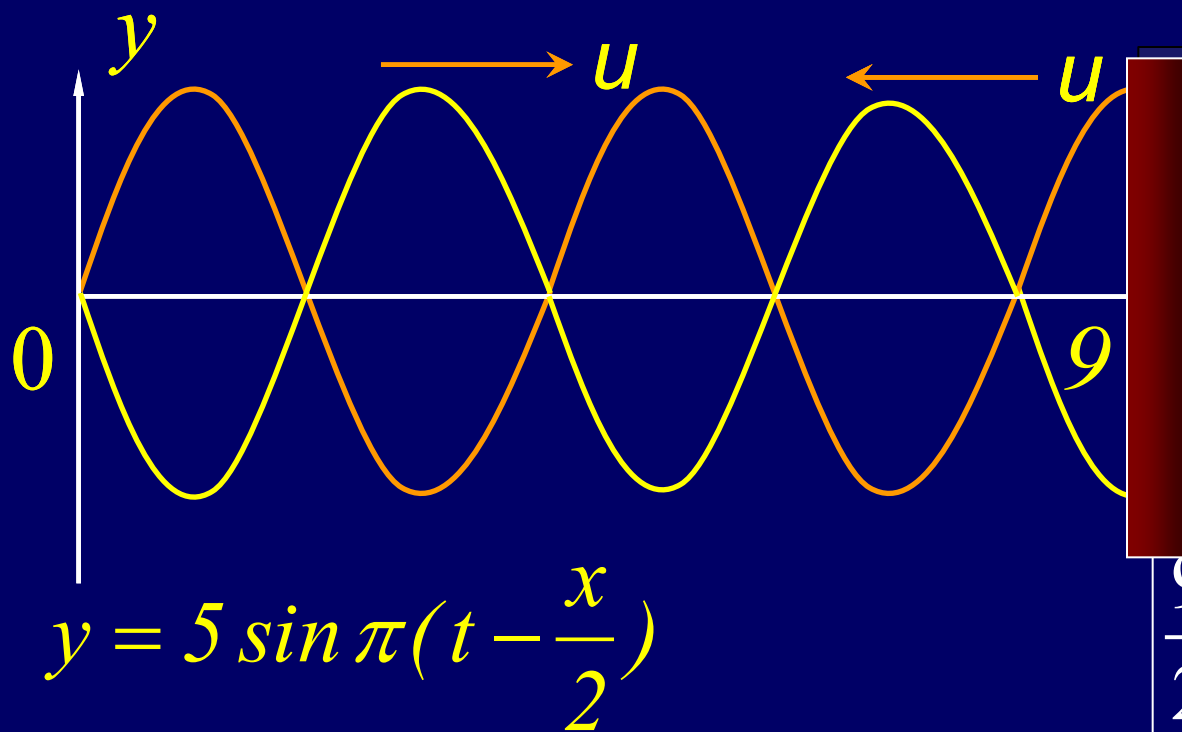
$$\frac{9}{2} \pi + \phi' = -\frac{\pi}{2}$$

$$\phi' = -5\pi$$

or

$$\phi' = \pi$$

$$y = 5 \sin \left[\pi \left(t + \frac{x}{2} \right) + \pi \right]$$



Example

Hard reflection

phase delay

$$\frac{9}{2}\pi + \phi' + \pi = -\frac{\pi}{2}$$

Reflection Wave

$$y' = 5 \sin[\pi(t + \frac{x}{2}) + \phi']$$

$$y' = 5 \sin[\pi(t + \frac{x}{2}) - 6\pi] = 5 \sin[\pi(t + \frac{x}{2})]$$

$$y + y' = 5 \sin[\pi(t - \frac{x}{2})] + 5 \sin[\pi(t + \frac{x}{2})] = 10 \sin \pi t \cos \frac{\pi x}{2}$$

Wave Function $y = y_m \sin[\omega(t - \frac{x}{u})]$

Reflection Wave $y = y_m \sin[\omega(t + \frac{x}{u}) + \phi']$

$$\phi' = -(\omega \frac{2x_0}{u} + \phi)$$

Soft reflection No phase delay $\phi = 0$

Hard reflection

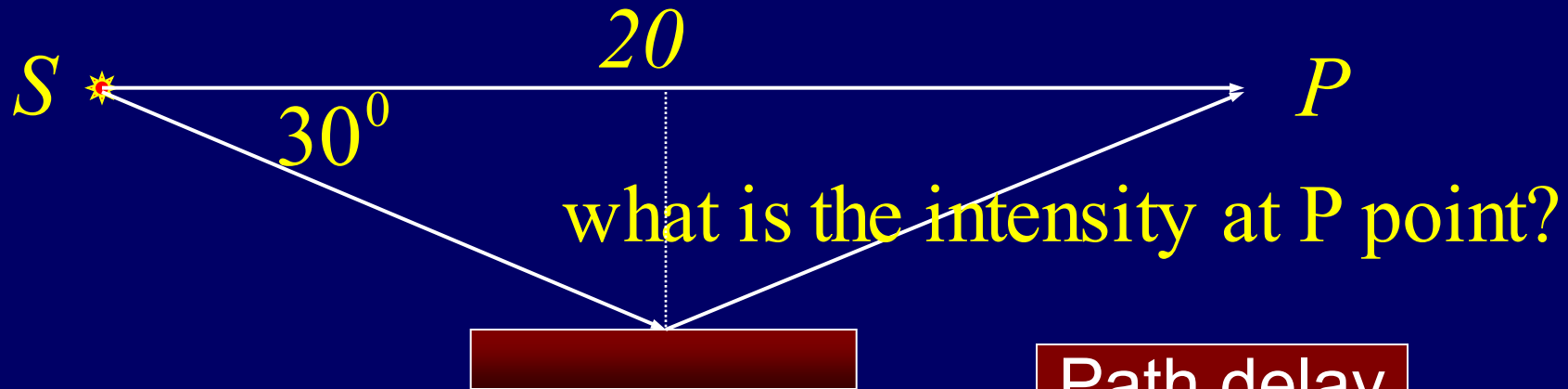
phase delay

$$\phi = \pi$$

$$\Rightarrow \Delta x = \frac{\lambda}{2}$$

path delay
半波损失

Example



Path delay

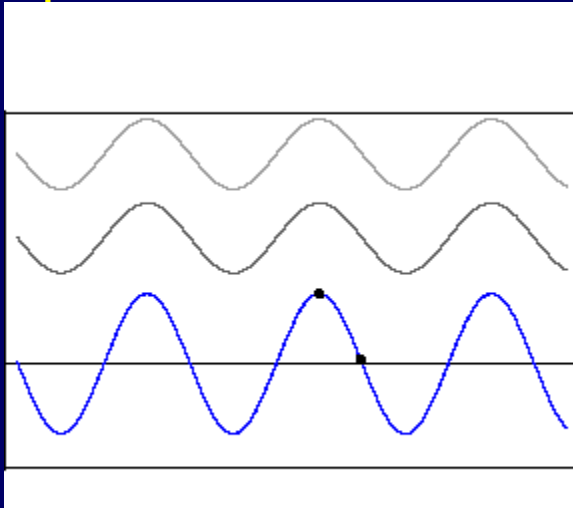
$$x_1 - x_2 = n\lambda \quad x_1 - x_2 = 2 \times 10 \times \frac{2}{\sqrt{3}} - 20 + \frac{\lambda}{2} \approx 4$$

$$x_1 - x_2 = \left(n + \frac{1}{2}\right)\lambda$$

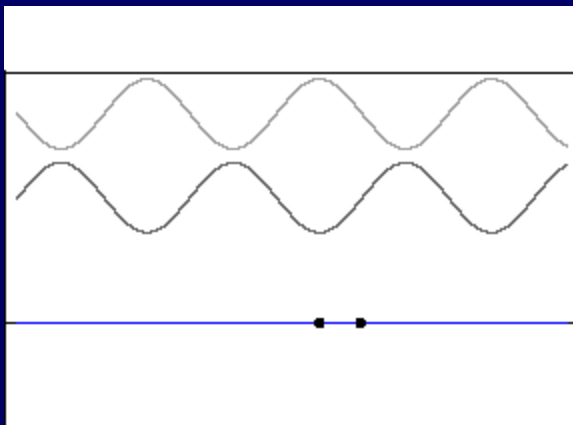
Max

Standing waves on a string

- Superposition of two waves moving in the same direction



- Superposition of two waves moving in the opposite direction

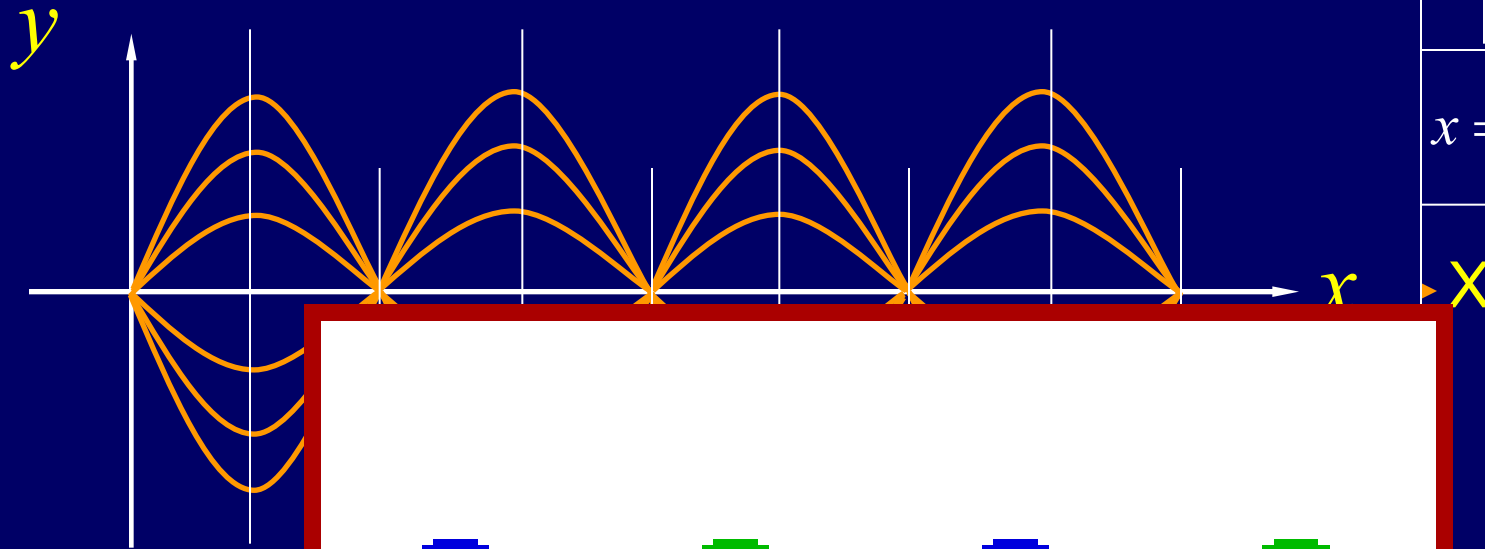


Standing Wave

$$y_1 = y_m \sin \omega t$$

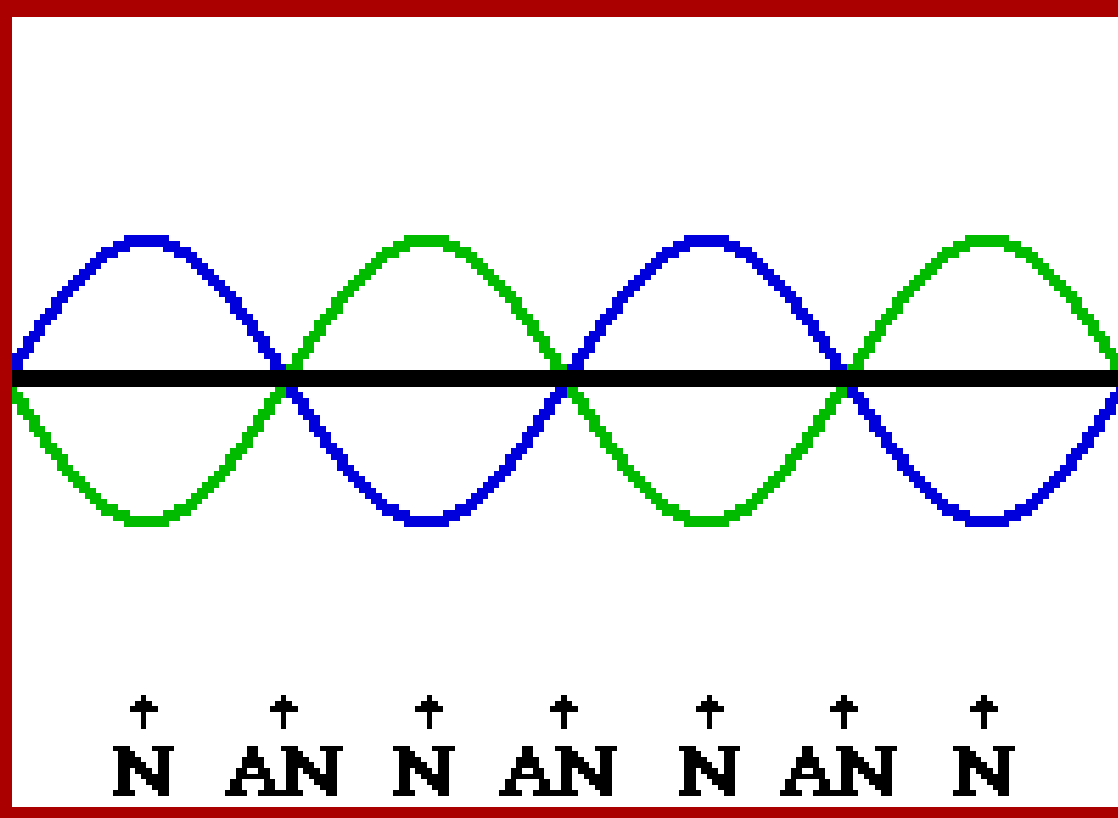
$$x = \frac{n\lambda}{2} \text{ max}$$

$$x = (n + \frac{1}{2}) \frac{\lambda}{2} \text{ min}$$



$$y = y_m \sin \omega t$$

$$y = [2y_m \sin \frac{x}{\lambda}] \sin \omega t$$



$$y(t + \frac{x}{u})]$$

$$\frac{x}{\lambda}] \sin \omega t$$

Normal modes of a string

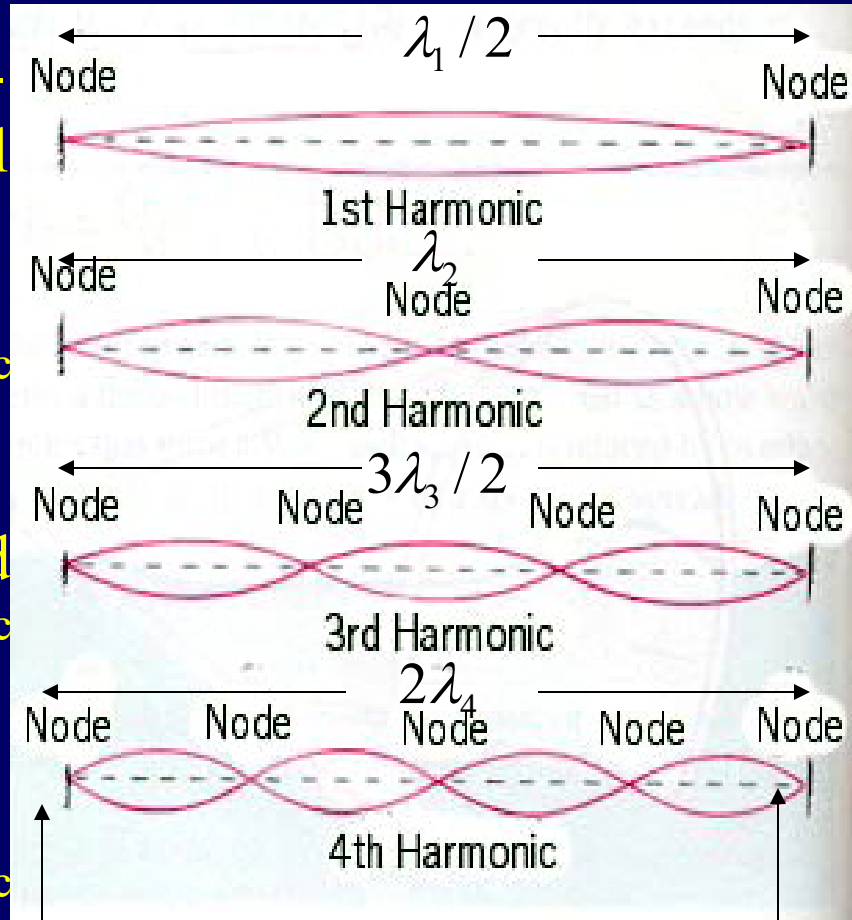
- There are infinite numbers of modes of standing waves

fundamental

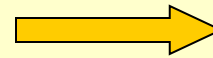
first harmonic

second harmonic

third harmonic



$$L = n \frac{\lambda}{2} \quad (n = 1, 2, 3, \dots)$$



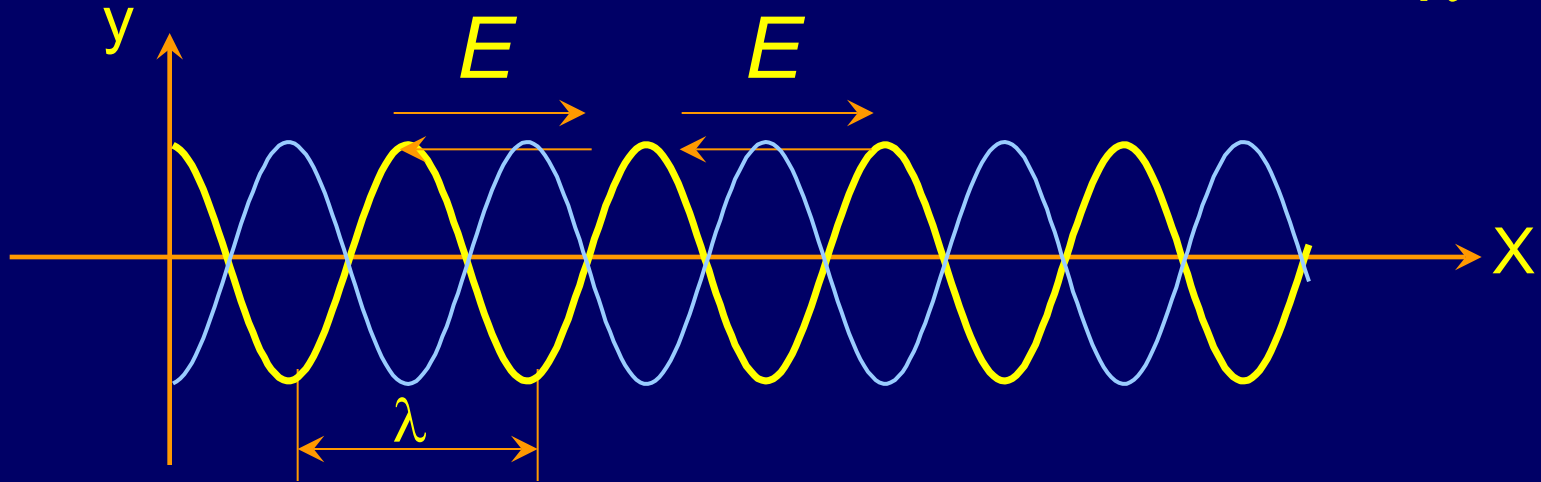
$$\lambda_n = 2L / n$$

$$f_n = n \frac{v}{2L} = f_1 = \frac{1}{2L} \sqrt{\frac{F}{\mu}}$$

fixed end L fixed end

Standing Wave

$$y = \left[2y_m \cos 2\pi \frac{x}{\lambda} \right] \sin \omega t$$

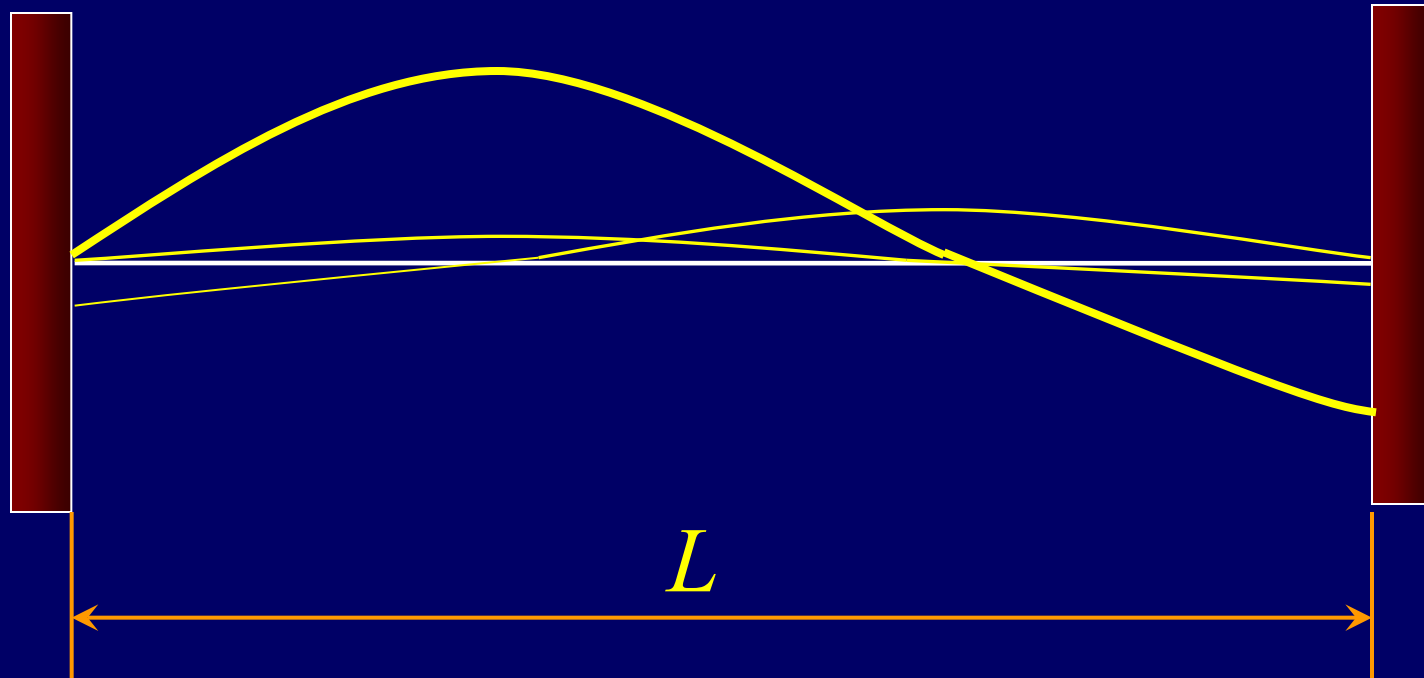


$$E = \frac{\mu y_m^2 \omega^2 \lambda}{16}$$

Total energy between N and AN
remains constant.

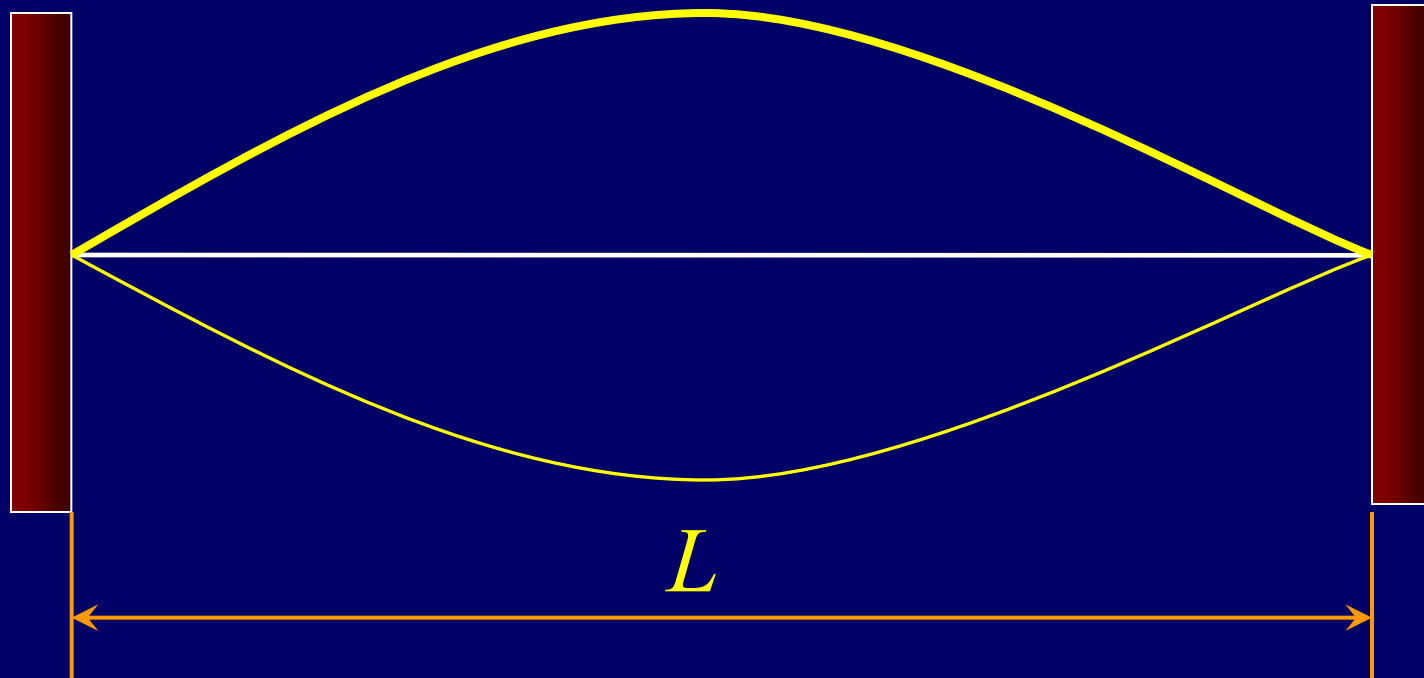
There is no power transmitted through a node or an antinode.

Resonance



Amplitude reduces

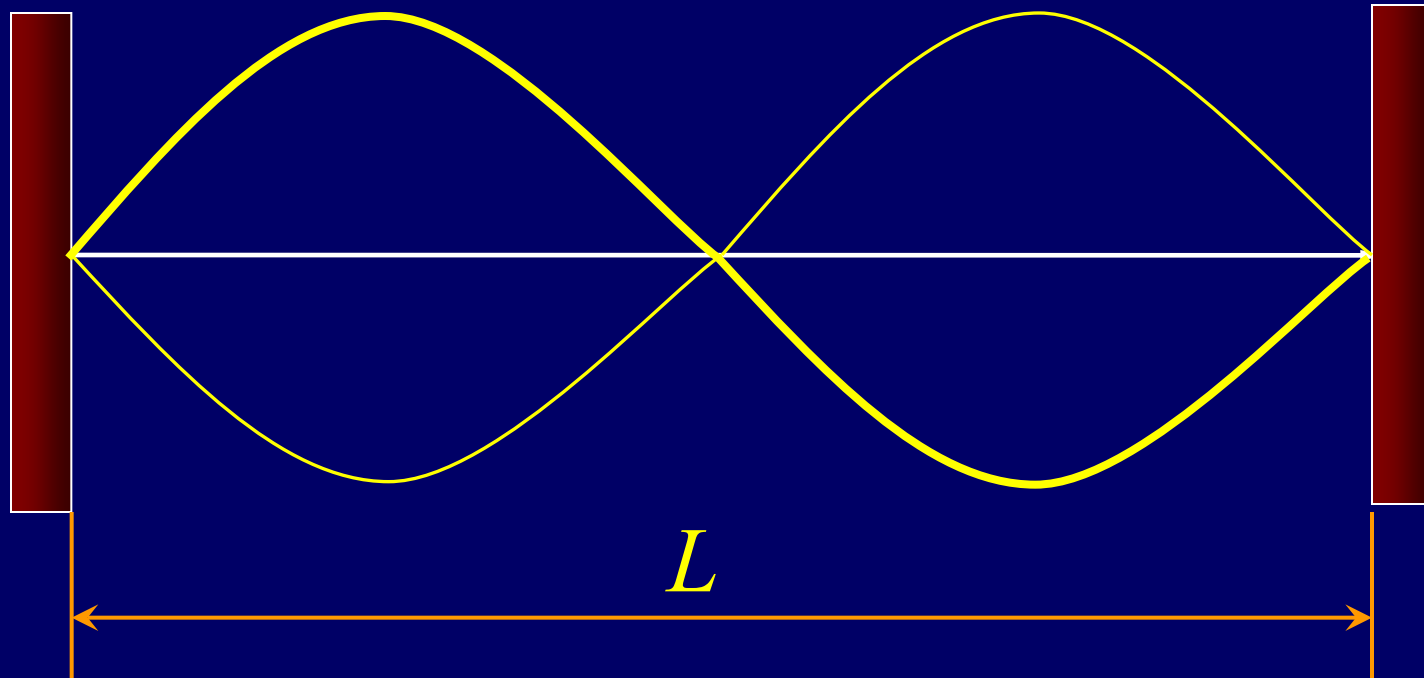
Resonance



$$L = \frac{\lambda}{2}$$

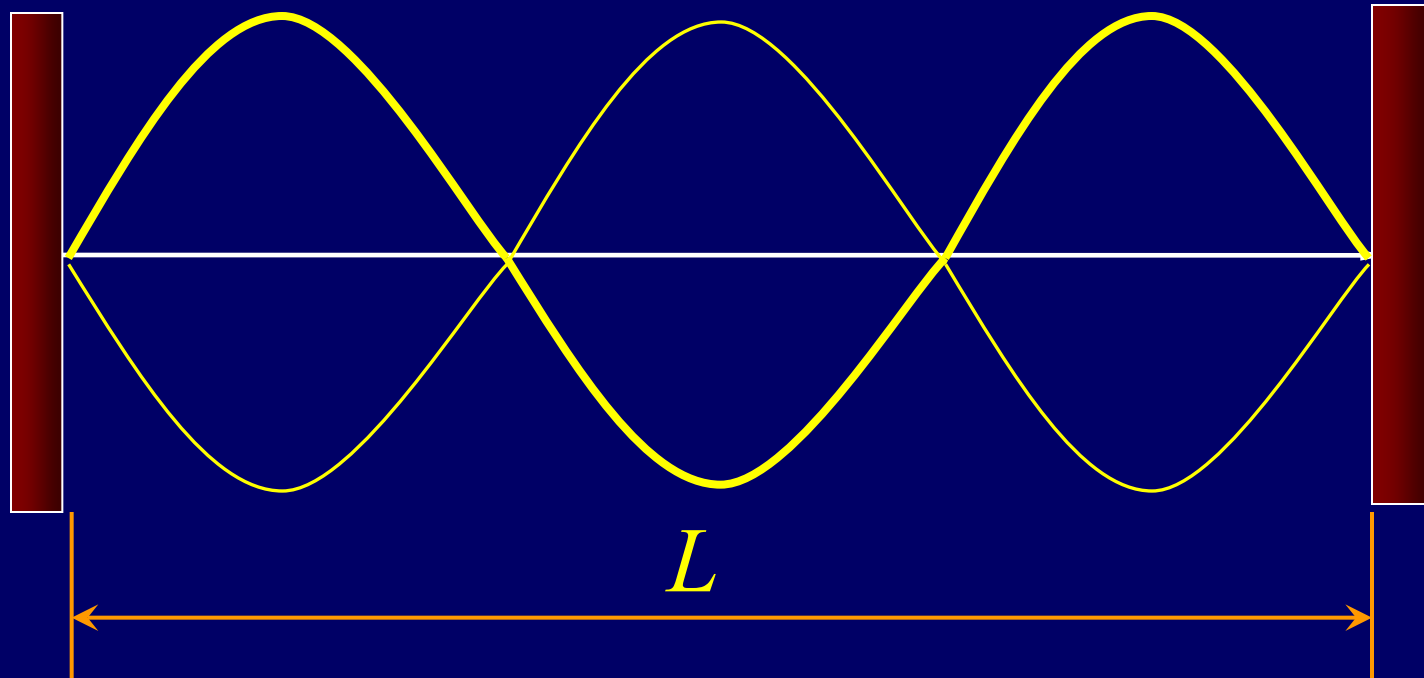
Amplitude does not reduce

Resonance



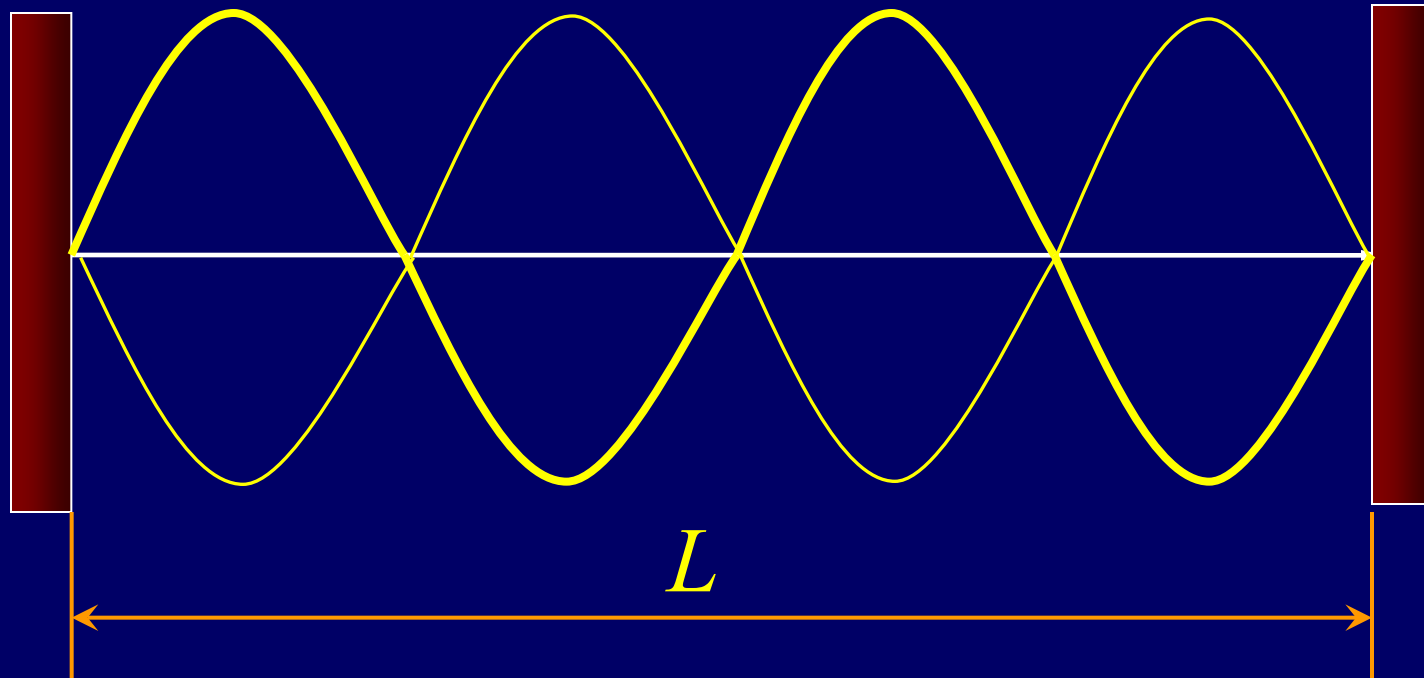
$$L = \lambda$$

Resonance



$$L = \frac{3\lambda}{2}$$

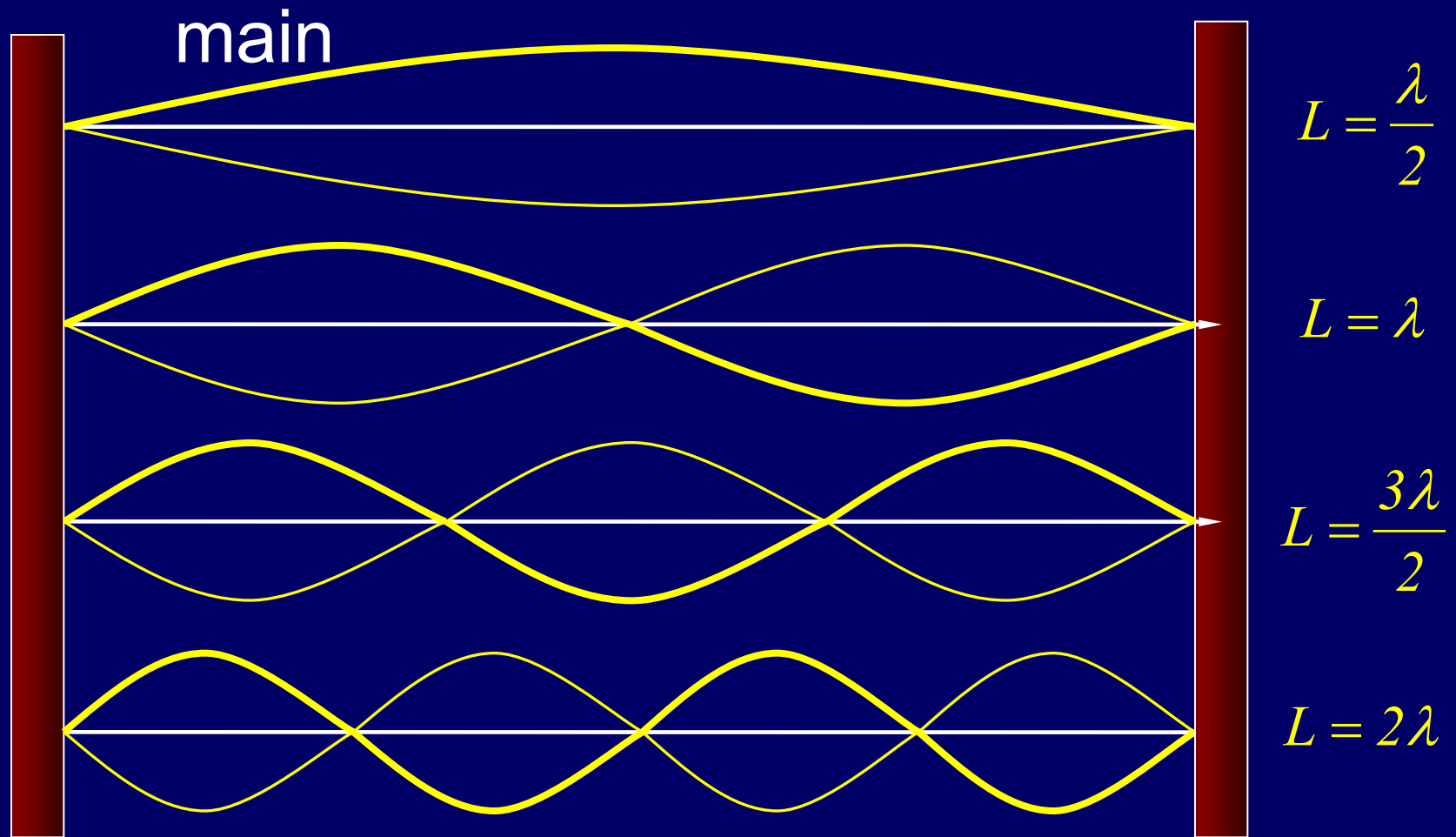
Resonance



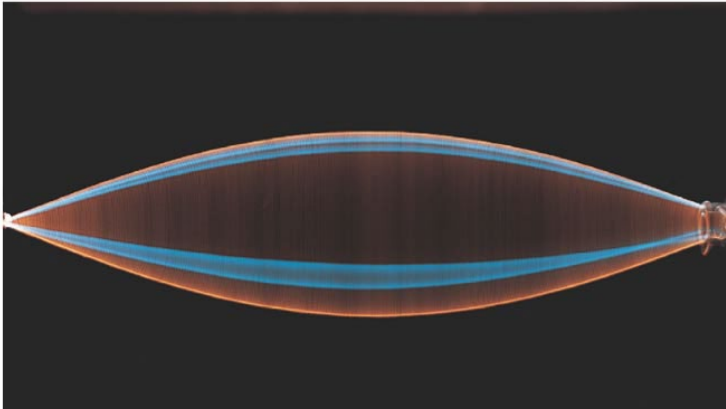
$$L = 2\lambda$$

Resonance

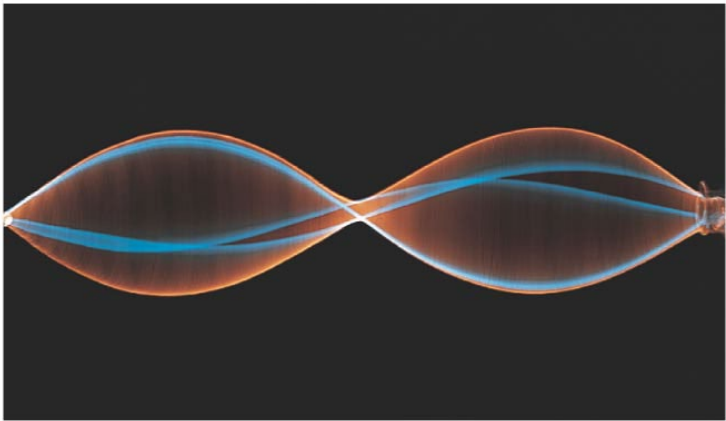
$$L = n \frac{\lambda}{2} \quad (n = 1, 2, 3, 4 \dots)$$



(a) String is one-half wavelength long.

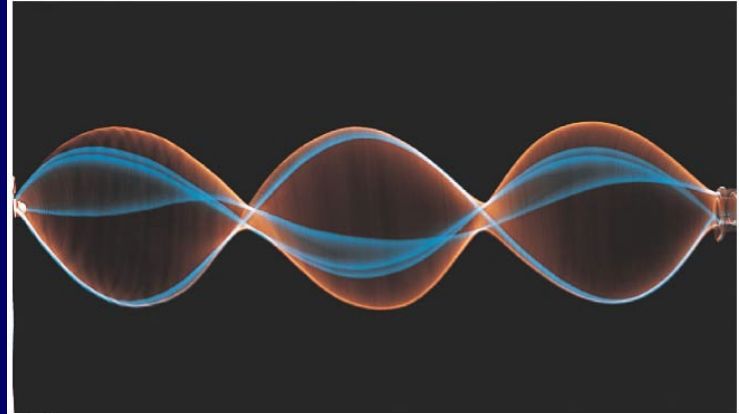


(b) String is one wavelength long.

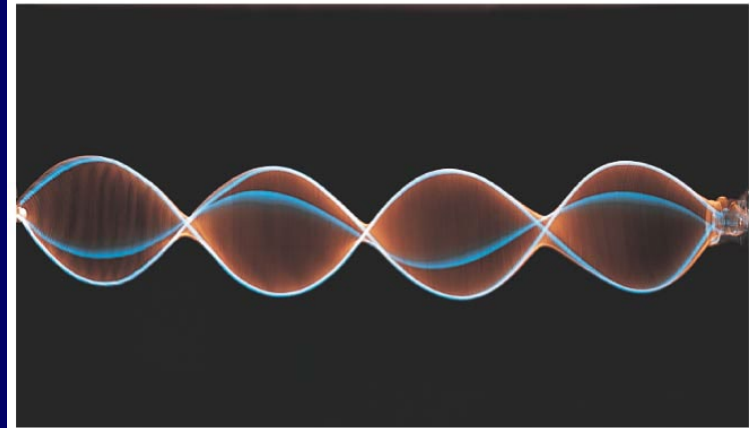


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(c) String is one and a half wavelengths long.



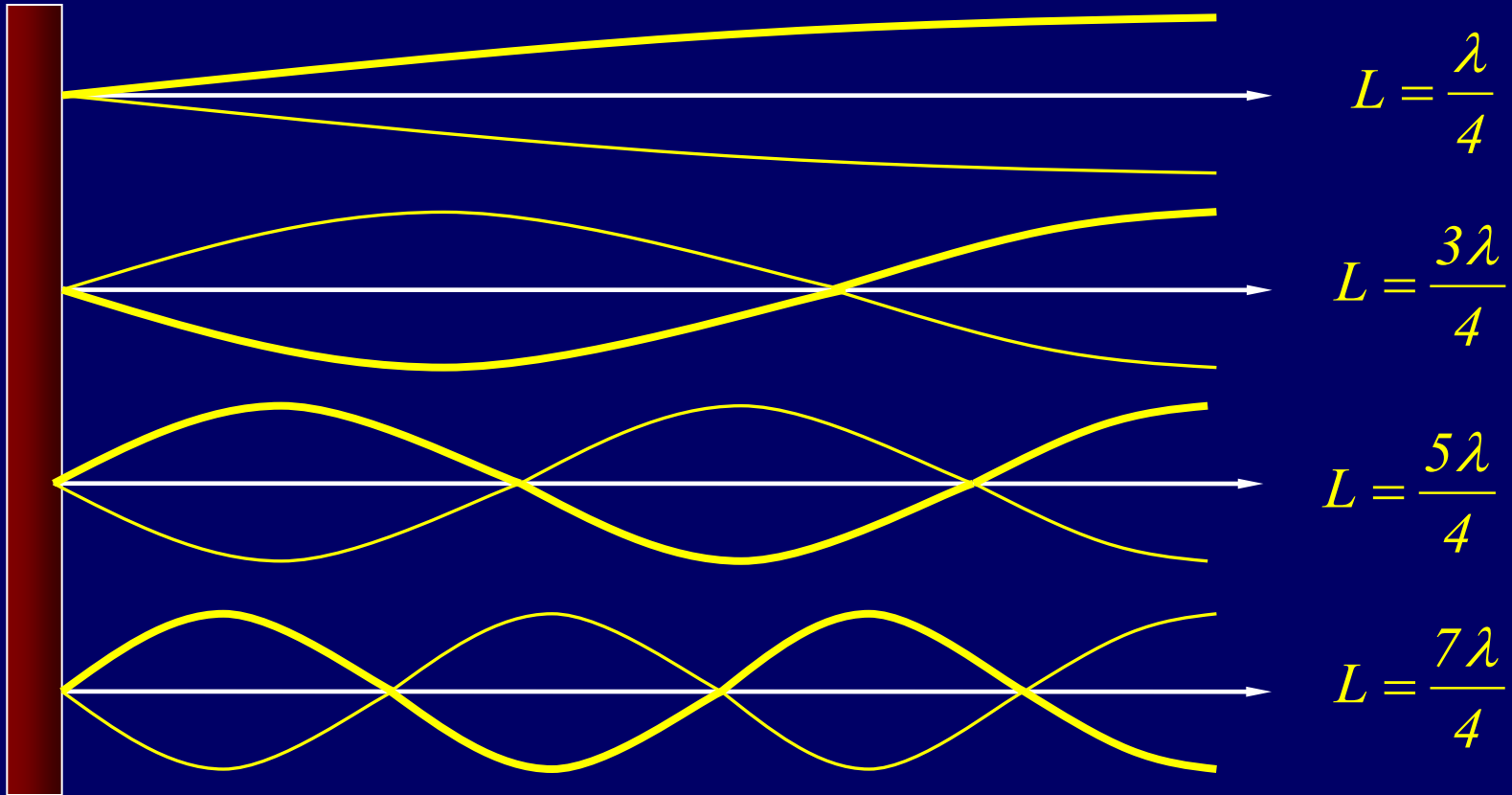
(d) String is two wavelengths long.



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Resonance

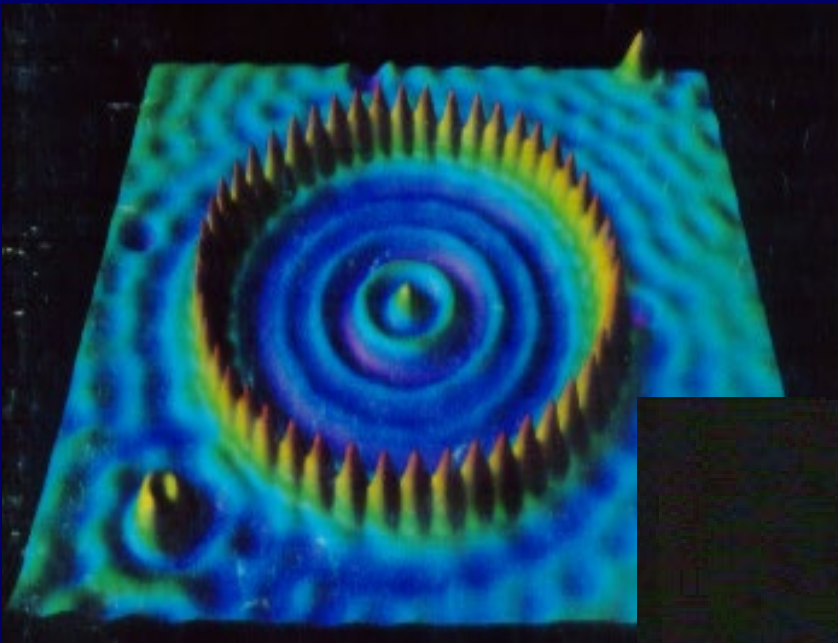
$$L = n \frac{\lambda}{4} \quad (n = 1, 3, 5 \dots)$$

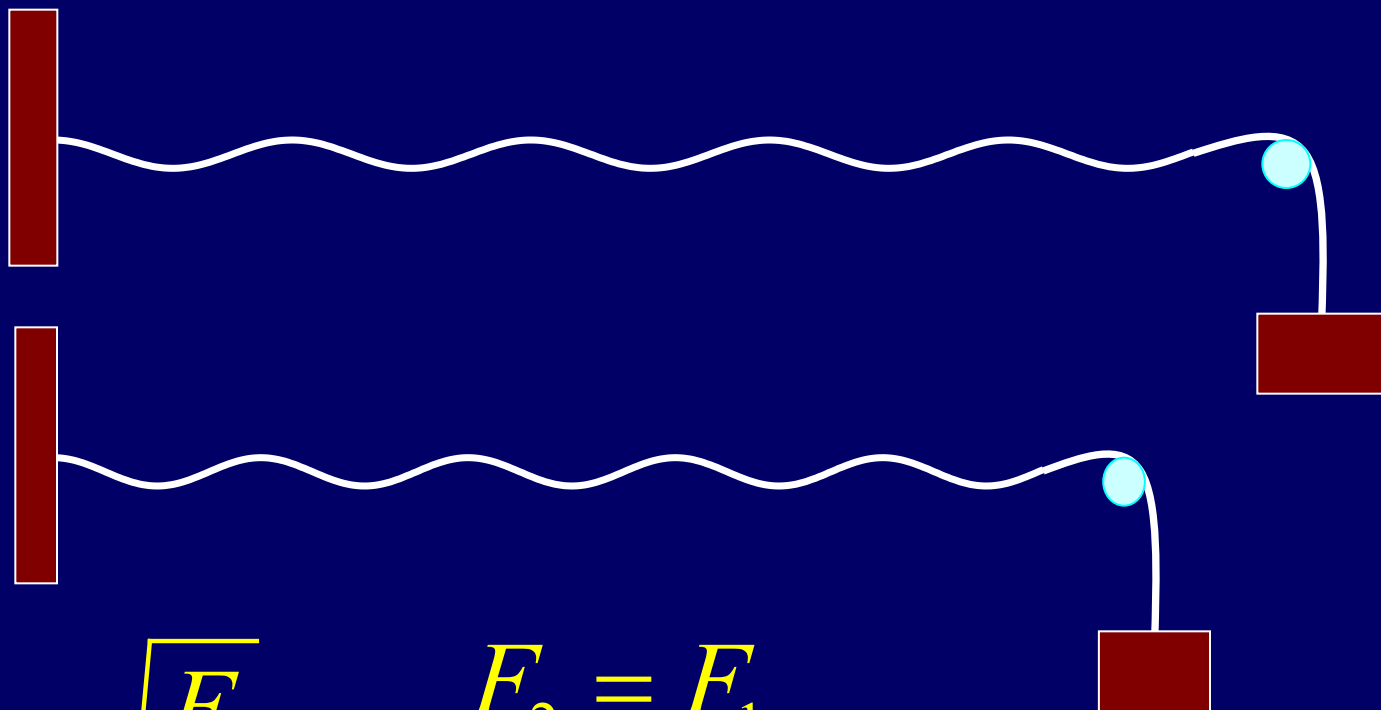


$$y + y' = 5 \sin\left[\pi\left(t - \frac{x}{2}\right)\right] + 5 \sin\left[\pi\left(t + \frac{x}{2}\right) + \pi\right] = 10 \cos(\pi t) \sin\left(\frac{\pi x}{2}\right)$$

where $x=9$ at the end

2-D Resonance





$$u = \sqrt{\frac{F}{\mu}}$$

$$F_2 = F_1$$

$$u_2 = u_1$$

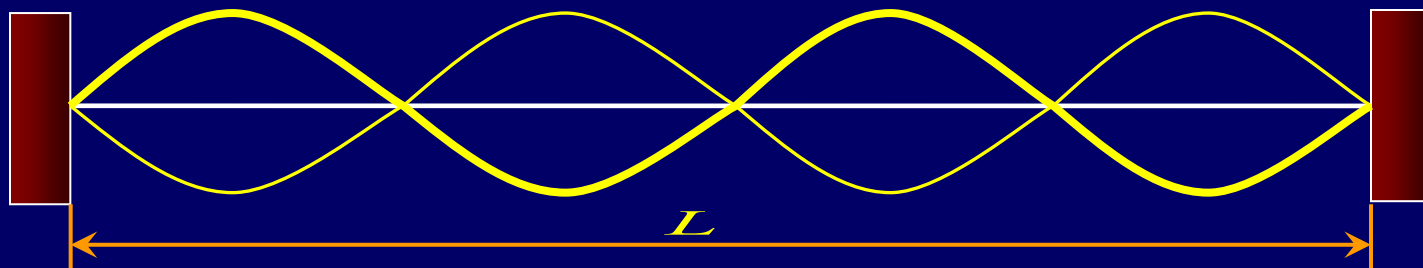
$$\lambda_2 < \lambda_1$$

$$f = \frac{u}{\lambda}$$

$$f_2 > f_1$$

Resonance

$$L = n \frac{\lambda}{2}$$



Resonance $L = n \frac{\lambda}{2} \quad \lambda = \frac{2L}{n} \quad u = \sqrt{\frac{F}{\mu}} \quad \text{string}$

$$F = \mu \cdot u^2 = \mu \cdot (\lambda f)^2 = \frac{4L^2 f^2 \mu}{n^2} \quad f = \sqrt{\frac{n^2 F}{4L^2 \mu}}$$

$$f_3 = \sqrt{\frac{9F}{4L^2 \mu}} \quad f_2 = \sqrt{\frac{F}{L^2 \mu}} \quad f_1 = \sqrt{\frac{F}{4L^2 \mu}}$$

Example

The length of a string, with density of mass $\mu = 0.02 \text{ Kg}$, is 0.4 m . If the tension force is 200 N . To give the frequency of the string.

$$u = \sqrt{\frac{F}{\mu}} = \sqrt{\frac{200}{0.02}} = 100 (\text{m} / \text{s})$$

$$L = n \frac{\lambda}{2} \quad \lambda = \frac{2L}{n}$$

$$f_n = \frac{u}{\lambda} = \frac{nu}{2L} = \frac{100n}{2 \times 0.4} = 125n$$

$$f_1 = 125 \text{ Hz} \quad f_2 = 250 \text{ Hz} \quad f_3 = 375 \text{ Hz}$$

main

CHAP. 18

Exercises

P421 2, 4, 7, 11, 12, 27, 32

Problems

P424 13, 14, 15